2026

CACS International Automatic Control Conference

暨國科會工程處控制學門114年度成果報告發表會

大會手冊

CACS 2026

2026 International Automatic Control Conference

Taichung, Taiwan

October 21-24, 2026

Final Program

Organized By

逢甲大學
Feng Chia University

Feng Chia University

Chinese Automatic Control Society

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Table of Contents

Welcome Messages from General Chairs 1

Committees 2

Special Sessions Organizers 6

Acknowledgment of Reviewers 10

Plenary Speeches 13

- **Plenary Speech I** 13
- **Plenary Speech II** 15
- **Plenary Speech III** 17

Conference Venue Information 19

Floor Plan 21

Transportation Information 22

Welcome Reception & Banquet Venue Details 27

Accommodation Information 29

Industrial Session & Exhibition 32

- **Industrial Session** 32
- **Exhibition** 32

Instructions for Physical Oral presentations 33

CACS 2026 Program Overview 34

Program at a Glance 36

CACS Technical Session Details 40

- **Session Track 1** 40
- **Session Track 2** 48
- **Session Track 3** 56
- **Session Track 4** 64
- **Session Track 5** 74
- **Session Track 6** 83

Welcome Messages from General Chairs

The International Automatic Control Conference (CACS 2026) is to be organized by the Chinese Automatic Control Society (CACS) and Feng Chia University (FCU) on Oct 21-24, 2026. The conference will take place in FCU ,Taichung City, TAIWAN.

This event provides scientists, engineers, and practitioners a great opportunity to present the latest design concepts, research results, developments, and applications in the fieId of automatic control and facilitates interactions between scholars and practitioners. Special sessions are to be organized by experienced researchers in the fieIds of automatic control, robotics, automation, artificial intelligence, smart vehicles, etc. to broaden the scope and promote participation. Accepted papers will be submitted for inclusion into IEEE Xplore subject to meeting IEEE Xplore's scope and quality requirements. Participants are also welcome to participate the Best Paper Contests.

CACS 2026 received a total of 297 submissions. At least three reviewers reviewed each paper to ensure thorough evaluation. Following the peer review process, 259 papers were accepted for presentation. Selected full papers presented at CACS 2026 will be published in the IEEE Xplore Digital Library. We would like to take this opportunity to express our sincere gratitude to all the reviewers of CACS 2026 for their dedicated and professional services. Participants are from nearly 12 different countries worldwide, thereby making CACS 2026 a truly international event.

Finally, we would like to thank all the Organizing Committee members, speakers, authors, and participants for their outstanding contribution and support to make CACS 2026 a success. We wish your CACS 2026 experience to be academically beneficial and enjoyable!

Yu-Chen Lin
General Chair, President of CACS

Feng Chia University,
Taiwan

Yo-Ping Huang,
General Chair, President of CACS

National Taipei University of Technology,
Taiwan

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國科會成果發表 NSTC Research Presentation

• Shin-Hung Chang, Chung Yuan Christian University, Taiwan
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• Hsu-Chih Huang, National Chung Cheng University, Taiwan
• Tsu-Wang Shen, Feng Chia University, Taiwan

Special Sessions Organizers

List of Special Sessions:

SS-01: Smart System and Its Applications, Prof. Yung-Yao Chen, Prof. Yu-Hsiu Lin, Prof. Sin-Ye Jhong

SS-02: Agentic and Physical AI-Driven Green Energy Transformation toward Net-Zero Energy Systems, Prof. Chun-Wei Huang

SS-03: Next-Generation Robotic Perception: Integrating Deep Learning and Machine Vision, Prof. Yi-Zeng Hsieh

SS-04: Data-driven Computational Intelligence Techniques and Applications, Prof. Chia-Feng Juang, Prof. I-Fang Chung

SS-05: Intelligent Mobile Robotic Systems and Applications, Prof. Ching-Chih Tsai

SS-06: Development of intelligent hearing aids and heart rhythm care devices, Prof. Cheng-Yuan Chang

SS-07: Automatic Control for Human-Robot Collaboration, Prof. Po-Ting Lin, Prof. Yung-Yao Chen

SS-08: Computational Intelligence Methods and Explainable AI for Evaluation and Distillation of AI Data Sets, Prof. Chen-Chia Chuang, Prof. Shun-Feng Su

SS-09: AI Based Robotics, Prof. Wen-June Wang

SS-10: Intelligent Control and Applications, Prof. Gwo-Ruey Yu

SS-11: Artificial Intelligence-Based Preclinical and Clinical Research on Alzheimer's Disease, Epileptic EEG, and Noninvasive Brain Stimulation, Prof. Po-Lei Lee

SS-12: AI-Driven Perception, Prediction, and Decision-Making for Autonomous Systems, Prof. Yu-Chen Lin

SS-13: Evolutionary Computation and Deep Learning for Autonomous Perception and Decision-Making, Prof Jia Guo, Prof. Jia-cheng Li, Prof. Nobuo Kawaguchi, Prof. Masato Noto, Prof. Bin Zhang, Prof. Akiyoshi Masanori

SS-14: System Integration and Control, Prof. Fu-Cheng Wang

SS-15: Applications of Generative Artificial Intelligence on Robotics, Prof. Kuu-Young Young, Prof. Ching-Hung Lee

SS-16: Computer Vision for Intelligent Robot Applications, Prof. Chung-Hsien Kuo, Prof. Huei-Yung Lin

SS-17: Data-Driven Control and Learning for Autonomous Mobile Systems, Prof. Hsin-Han Chiang, Prof. Chia-Wen Chang

SS-18: AI-Empowered Smart Driving Technology, Prof. Yen-Lin Chen, Prof. Hsin-Han Chiang

SS-19: Intelligent Control in Robotics and Energy Applications, Prof. Sei Zhen Khong, Prof. Kai-Yi Wong

SS-20: Multimodal AI, Computer Vision, and Intelligent Systems for Real-World Applications, Prof. Sze-Teng Liong, Prof. Tsung-Yi Chen

SS-21: Robotic manipulator technology and applications, Prof. Syuan-Yi Chen, Prof. Jeng-Dao Lee

SS-22: Robust Quantum Filter and Control Designand Their Applications to Quantum Processing and Communication, Prof. Bor-Sen Chen

SS-23: Development and Integration of Intelligent Medical Diagnosis and Rehabilitation Care Systems, Prof. Chih-Min Lin

SS-24: Integrated Monitoring and Control, Prof. Yo-Ping Huang, Prof. Yue-Shan Chang

SS-25: Adaptive and learning control of intelligent robotic systems, Prof. Feng-Li Lian, Prof. Ming-Li Chiang, Prof. Li-Chen Fu

SS-26: Advanced Guidance, Navigation and Control Systems for Unmanned Aircraft Systems, Prof. Ying-Chih Lai

SS-27: Real-Time Intelligent Systems and Control for Advanced Mechatronics, Prof. Nien-Tsu Hu, Prof. Chin-Sheng Chen, Prof. Ming-Hung Lin

SS-28: Designs and Applications of GAI based Service Robots, Prof. Tzuu-Hseng S. Li

SS-29: Observer-based control design for mechatronic systems, Prof Ming-Yang Cheng

SS-30: Perception and Decision-Making of Intelligent Systems, Prof. Chen-Chien Hsu, Prof. Wei-Yen Wang, Prof. Min-Jie Hsu, Prof. Cheng-Kai Lu, Prof. Wen-Lian Hsu

SS-31: Advanced Control and Analysis for Intelligent Mechatronic Systems, Prof. Shyh-Leh Chen

SS-32: BrainSynergy: An Open-Source AI Platform for Multimodal Neuroimaging and Applications Across the Lifespan, Prof. Jeen-Shing Wang

SS-33: Smart Robotics and Intelligent Systems Applications, Prof. Yu-Kun Lin, Prof. Chiu-Hung Chen

SS-34: Smart Home Service Robot, Prof. Ching-Chang Wong, Prof. Hsuan-Ming Feng, Prof. Chi-Yi Tsai

SS-35: Artificial Intelligence and Multi-Modality Imaging in Precision Radiation Therapy , Prof. Weir-Chiang Youm Prof. Yu-Te Wu, Prof. Kai-Lin Yang

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Ting-Wan Chang	Yih-Guang Leu	
Tsai-Fu Wu	Yin-Tien Wang	

Plenary Speeches

Plenary Speech I

Date/Time: October 22, 2026/ 11:10 ~ 12:00

Room: Virtuosi Hall Room V001

Talk Title: From Tokens to Topology: System-Level LLM Optimization for Latency, Throughput, and Cost in Production Workflows

Prof. Yiran Chen
Duke University, USA

Chair: Prof. Yo-Ping Huang

Biography:

Yiran Chen (Fellow, IEEE) received the Ph.D. degree from Purdue University in 2005. He is currently the John Cocke Distinguished Professor of electrical and computer engineering with Duke University, the Director of the NSF AI Institute for Edge Computing Leveraging the Next-Generation Networks (Athena) and the NSF Industry-University Cooperative Research Center (IUCRC) for Alternative Sustainable and Intelligent Computing (ASIC), and the Co-Director of the Duke Center for Computational Evolutionary Intelligence (DCEI). He received 11 best paper awards, one best poster award, and 15 best paper nominations from international

conferences and workshops. He received numerous awards for his technical contributions and professional services, such as the IEEE CASS Charles A. Desoer Technical Achievement Award and the IEEE Computer Society Edward J. McCluskey Technical Achievement Award. He has been the Distinguished Lecturer of IEEE CEDA and CAS.(Based on document published on 6 February 2025).

Plenary Speech II

Date/Time: October 22, 2026/ 13:10 ~ 14:00

Room: Virtuosi Hall Room V001

Talk Title: From Zero-Phase-Error-Tracking to Optimal Inversion to High-Performance Motion Control

Prof. Tsu-Chin Tsao
University of California, USA

Chair: Prof. Li-Wei Ko

Abstract:

Mechanical motion generation and vibration suppression is fundamental to modern machines and emerging innovations.Abilities to learn and compensate for complex mechanical system and disturbance dynamics are key to synthesizing adequate control actions to achieve precision motions, where dynamic inversion has a central role in synthesizing feedforward and feedback control. This talk will start from the simple and celebrated Zero-Phase-Error-Tracking- Control (ZPETC), to more general Optimal Dynamic Inversion, with a note on the optimality of ZPETC, and to high-performance control, synthesized by dynamic inversion, for addressing narrowband (repetitive control, iterative learning control) and broadband (adaptive control) motions and disturbances in real world applications.

Biography:

Dr. Tsu-Chin Tsao (B.S. National Taiwan University, Taiwan, M.S., Ph.D. U.C. Berkeley) is Distinguished Professor at the Mechanical and Aerospace Engineering Department, Samueli School of Engineering and Applied Science, University of California Los Angeles. His research interest includes control systems, mechatronics, and robotics. An ASME Fellow, Dr. Tsao has served in leadership and editorship positions as Department Chair and Journal's Editor-in-Chief. His work was recognized by several Best Paper Awards, International Federation of Automatic Control (IFAC) Mechatronic Systems Outstanding Research Award, ASME Henry M. Paynter Outstanding Investigator Award, and IFAC Outstanding Service Award. Dr. Tsao has engaged in numerous fundamental and applied research projects for science foundations, mission oriented agencies, and industrial sectors in manufacturing, automotive, aerospace, medical and health.

Plenary Speech III

Date/Time: October 23, 2026/ 09:00 ~ 10:00

Room: Virtuosi Hall Room V001

Talk Title: How Memory Powers Intelligence: From Cognition to AI Computing Systems

Prof. Hai Li
Duke University, USA

Chair: Prof. Chia-Feng Juang

Biography:

Hai (Helen) Li (Fellow, IEEE) received the Ph.D. degree from Purdue University in 2004. She is currently the Clare Boothe Luce Professor and the Department Chair of the Electrical and Computer Engineering Department, Duke University. Her current research interests include neuromorphic circuits and systems for brain-inspired computing, machine learning acceleration and trustworthy AI, conventional and emerging memory design and architecture, and software and hardware co-design. She was a recipient of the NSF Career Award in 2012, the DARPA Young Faculty Award in 2013, the TUM-IAS Hans Fischer Fellowship from Germany in 2017, and the ELATE Fellowship in 2020. She received nine best paper awards and additional nine best paper nominations from international conferences. She is a Distinguished Lecturer of the IEEE CAS Society from 2018 to 2019 and

a Distinguished Speaker of ACM from 2017 to 2020.(Based on document published on 6 February 2025).

Conference Venue Information

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逢甲大學商學大樓
逢甲夜市
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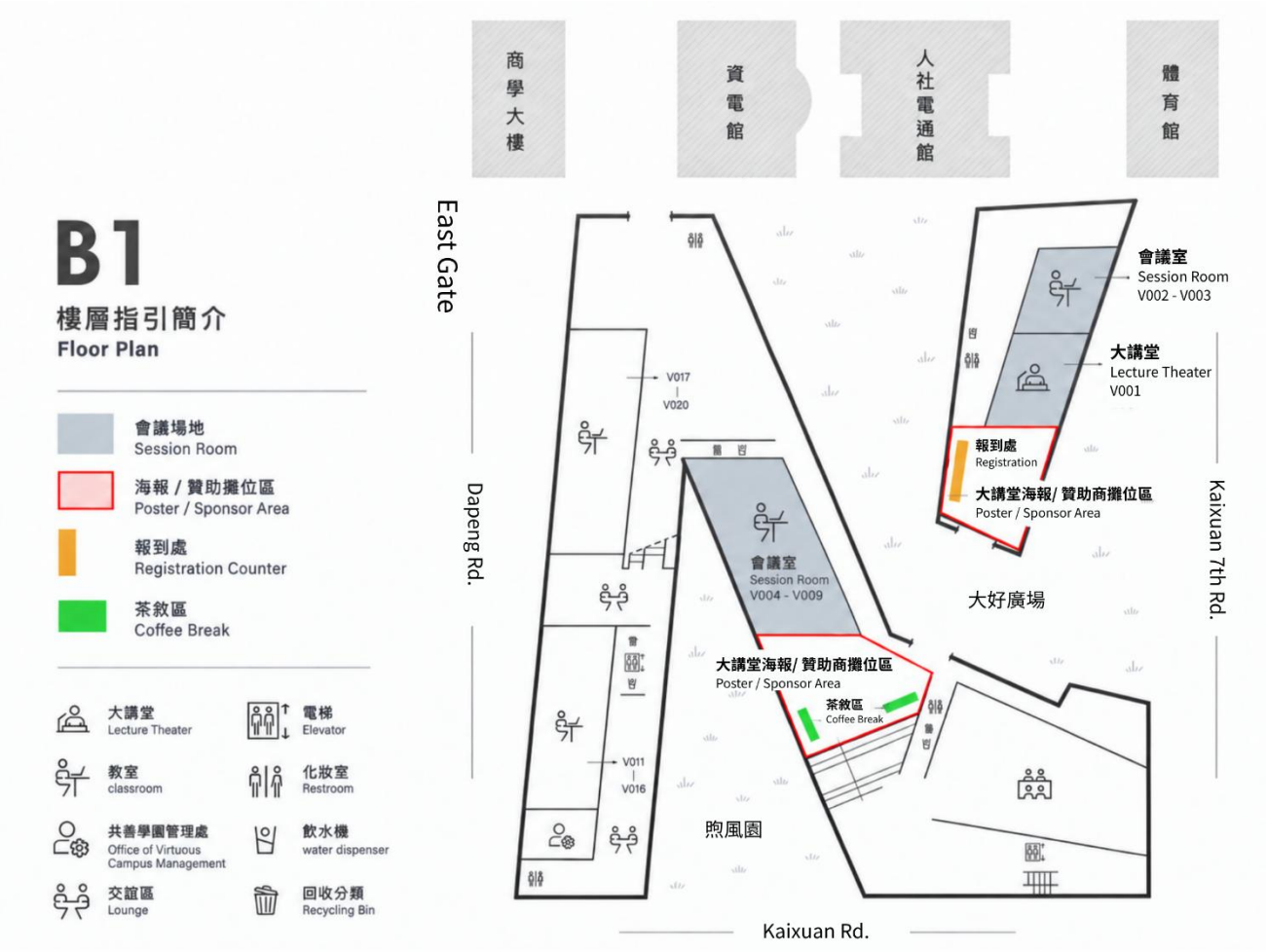
刁民-酸菜魚 逢甲店
HOME燒肉-逢甲店
彩碗ColorBowl Poke 西屯店
萬客什錦 青海店
炙肉高手-台中逢甲店
麥當勞-台中河南二餐廳-設有得來速
水滴中央公園 白色藝術牆
臺中中央球場
水滴經貿園區
中央公園遊客中心
極光情境旅館 AURORA MOTEL
浮雲客棧Kloud Hotel
星享道雅樂軒酒店
文修公園
福星公園
中國醫藥大學 質子治療中心
曆秘功夫菜手路湯甘肅店
LUCY'S CAF
文華路130巷
文華路86巷
凱旋七街
凱旋六街
凱旋三街
凱旋二街
大鵬路
河南路二段
中平路
福星路
逢甲路
西安街
福上巷
大河一巷5弄
永輝路75巷
大河街32巷

Floor Plan


B1
樓層指引簡介
Floor Plan
會議場地
Session Room
海報 / 贊助攤位區
Poster / Sponsor Area
報到處
Registration Counter
茶敘區
Coffee Break
大講堂
Lecture Theater
電梯
Elevator
教室
classroom
化妝室
Restroom
共善學園管理處
Office of Virtuous Campus Management
飲水機
water dispenser
交誼區
Lounge
回收分類
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體育館
East Gate
Dapeng Rd.
Kaixuan 7th Rd.
Kaixuan Rd.
V017
V020
V011
V016
會議室
Session Room
V004 - V009
大講堂海報/ 贊助商攤位區
Poster / Sponsor Area
茶敘區
Coffee Break
大好廣場
煦風園
會議室
Session Room
V002 - V003
大講堂
Lecture Theater
V001
報到處
Registration
大講堂海報/ 贊助商攤位區
Poster / Sponsor Area

Transportation Information

1. 2026 CACS Shuttle Bus

CACS 2026 provide free shuttle buses between the conference venues, with a detailed timetable listed below:

October 22, Thursday

THSR → FCU		FCU → THSR		FCU → Insky Hotel		Insky Hotel → THSR	
07:30	09:30	12:00		17:00		20:10	
08:00	11:00	17:30		17:20		20:40	
08:30	12:00						

October 23, Friday

THSR → FCU		FCU → THSR		FCU → Truelove Garden Hotel		Insky Hotel → THSR	
07:30	09:30	12:00		17:20		20:30	
08:00	11:00	17:30		17:30		21:00	
08:30	12:00			17:40		21:30	

Upon arrival at Taichung HSR Station, please proceed through the ticket gates and look for the **CACS 2026 reception staff**. They will guide you to the designated shuttle bus pick-up location.

2. Passenger transport

Take an intercity bus such as Ubus, Kuo-Kuang Motor Transport, or Ho-Hsin Bus and get off at either Zhonggang Station or Chaoma Station. From there, you can take a taxi (about 5-10 minutes) or transfer to a city bus to reach Feng Chia University.

- From Ubus Zhonggang Transfer Station, take bus 28 to the university.
- From Chaoma Station, take a taxi (about 5-10 minutes) or a city bus to the university

3. By train: Taichung Station

Please take the Western Trunk Line (Mountain Line) to Taichung Station. At the train station, exit via the [Taiwan Boulevard] exit and go to Platform D at the bus lane. Transfer to bus 5, 25, or 33, and get off at the Feng Chia University stop.

For bus information, please refer to the "Taichung City Bus Real-and relevant Taichung bus apps

4. By Taiwan High Speed Rail (THSR):Taichung Station

You can take a taxi via the Zhongzhang Expressway, which takes about 25 minutes to reach Feng Chia University. Alternatively, you can use the HSR transfer service: go to Exit 6 at Taichung HSR Station, proceed to Bus Platform 12, and take Bus 160 (HSR Taichung Station - Overseas Chinese University). Get off at the Feng Chia University stop.
Or you can take the Taichung MRT Green Line from HSR Taichung Station and get off at:

- Wenxin Yinghua Station
- Wenhua Senior High School Station

Then transfer to a city bus bound for Feng Chia University or Dapeng New Community

5. Drive to conference venue

- Main Campus (Main Gate) - No. 100, Wenhua Rd., Xitun Dist., Taichung City 407, Taiwan
- Main Campus (East Gate) - No. 127, Fengda Rd., Xitun Dist., Taichung City 407, Taiwan

2026 CACS provides four partner parking lots with complimentary parking for conference guests:

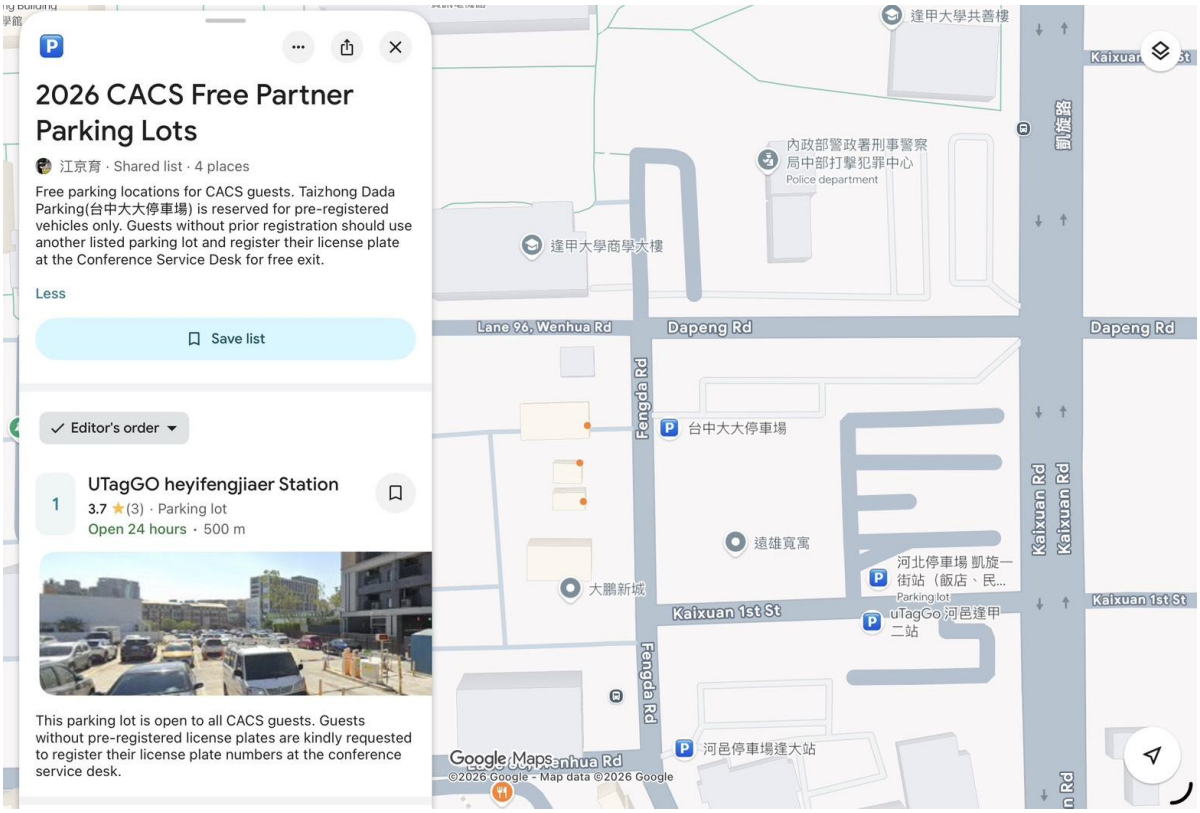
1. TagGO heyifengjiaer Station(uTagGo 河邑逢甲二站)

2. 河北停車場 凱旋一街站（飯店、民宿合作停車場）

3. Heyitingchechangfengda Station(河邑停車場逢大站)

4. Taizhong Dada Parking(台中大大停車場)

Parking Notice:
Guests with pre-registered license plates may park at any of the four partner parking lots.
Please note that Taizhong Dada Parking is available only to guests with pre-registered license plates. Guests without prior license plate registration should use one of the other three partner parking lots.
Guests without prior registration who use the other designated parking lots should register their license plate number at the Conference Service Desk before leaving to receive complimentary parking.



Google Maps – 2026 CACS Free Partner Parking Lots

https://maps.app.goo.gl/Ua59N2ECs4XZjQ6n6?g_st=il

2026 CACS Free Partner Parking Lots_QRCODE

Welcome Reception & Banquet Venue Details

Welcome Reception, Thursday, October 22, 2026

Aloft Taichung

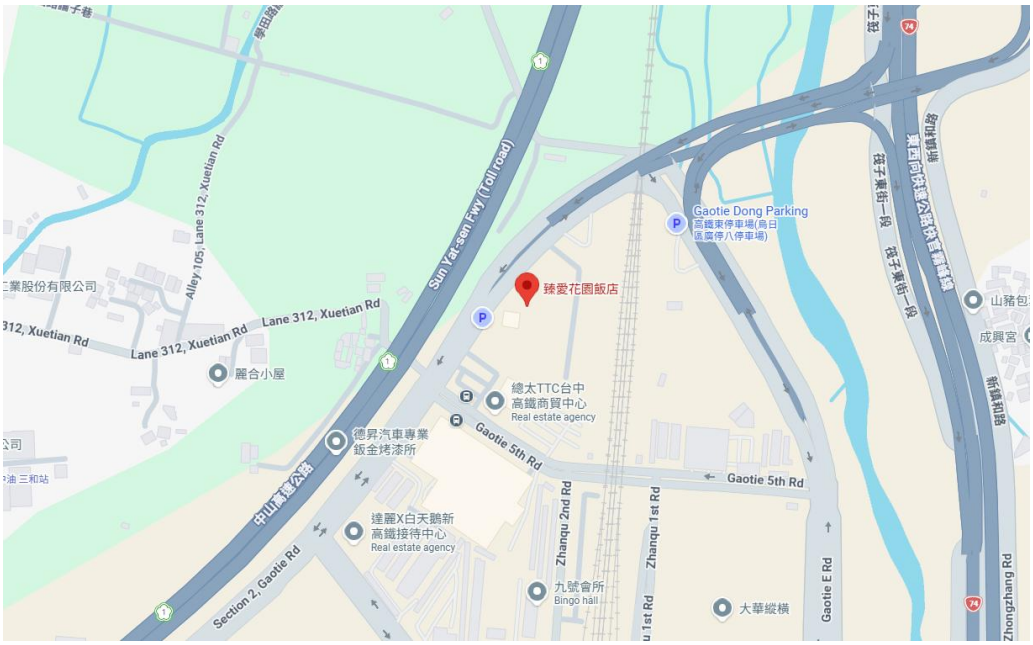
Address: No. 18, Fuxing N Rd, Xitun District, Xitun District,Taichung, Taiwan
Tel : 886 4-27083308

Banquet Dinner Venue

Conference Banquet, Friday, October 23, 2026

Truelove Garden Wuri, Taichung

Address: No. 168, Sec. 3, Gaotie Rd., Wuri Dist., Taichung City, Taiwan
Tel : 886 4-2338-2669



Accommodation Information

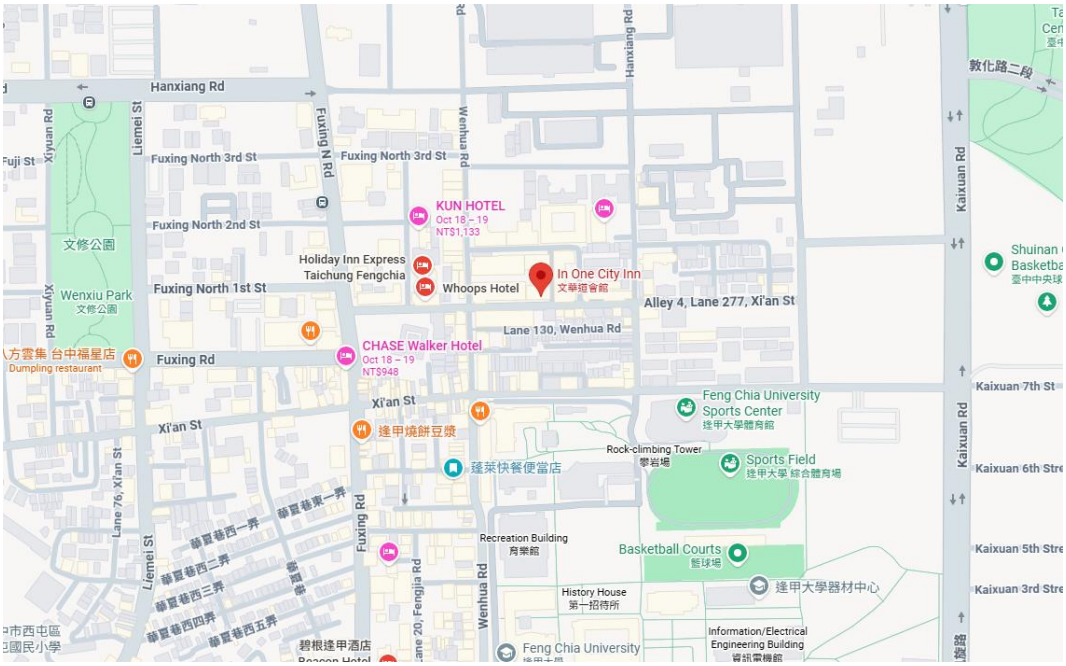
Insky Hotel (星享道酒店)

Add No. 18, Fuxing N Rd, Xiping Village, Xitun District, Taichung City, 407, Taiwan
E-Mail : reservation@inskyhotel.com Tel : 04-2706-3608

In One City Inn (文華道會館)

Add No. 31, Ln. 138, Wenhua Rd., Xiping Vil., Xitun Dist., Taichung City 407, Taiwan

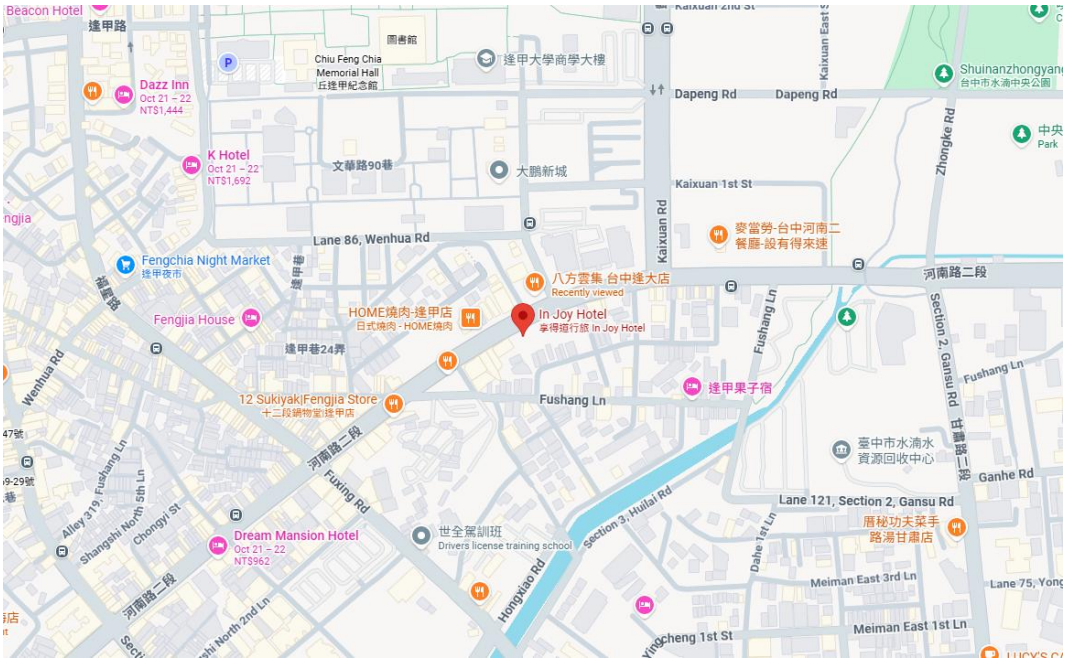
E-Mail : reservation@inone.com.tw Tel : 04-3602-8088



In Joy Hotel (享得道行旅)

Add No. 211, Sec. 2, Henan Rd., Fengfu Vil., Xitun Dist., Taichung City 407, Taiwan

E-Mail : reservation@injoyhotel.com.tw Tel : 04-2708-0189



Industrial Session & Exhibition

Industrial Session

Date/Time : October 22, / 14:00 ~ 15:15

Location : Virtuosi Hall

Time	Corporation
14:00 ~ 14:15	台達電子工業股份有限公司
14:15 ~ 14:30	昊青股份有限公司
14:30 ~ 14:45	虎門科技股份有限公司
14:45 ~ 15:00	鈦思科技股份有限公司
15:00 ~ 15:15	誼卡科技顧問股份有限公司

Exhibition

Date/Time : October 22, / 08:00 ~ 17:00
Date/Time : October 23, / 08:00 ~ 17:00

Location : Virtuosi Hall

PS: Exhibitors are requested to complete dismantling by 5:00 p.m. on October 23

Instructions for Physical Oral presentations

1. The time slot for each presentation is 12 minutes, plus 3 minutes for Q&A. Please adjust your speech length and the number of slides to stay within the allowed time limit. Session chairs will strictly enforce the time limits of the schedule.
2. Please pre-load your presentation file(s) onto the conference laptop before 10 minutes of the session starting time at the allocated hall for your presentation. Here, we request physical presenters to use our laptop to make the presentation so that the virtual part can run smoothly.
3. Please rename your file as follows: #paperID_first name_last name(for example: #1000_CACS_FCU). Presentations prepared in Microsoft PowerPoint format (.ppt, pptx) is recommended.
4. Confirm the time allocated for your presentation from the session schedule (https://cacs2026.github.io/cacs2026-program/).
5. Please make your presence known to one of the session chairs before your session starts. Please arrive at the session room at least 10 minutes ahead of the starting time of each session and sit in a front row to lessen transition time between presentations.
6. Please do not shift the presentation order or time slot without the session chairs' agreement.
7. Free Wi-Fi is available at the conference venue:
   Location: Virtuosi Hall
   Wi-Fi Name: 2026CACS
   Password: (None)

CACS 2026 Program Overview

Wednesday, October 21, 2026

Time	Topic	Venue
08:00~17:00	Registration	Virtuosi Hall Conference Lobby

Thursday, October 22, 2026

Time	Topic	Venue
08:00~17:00	Registration & Exhibition	Virtuosi Hall Conference Lobby
09:00~10:30	Sessions Track 1 / NSTC Poster 1	Session Rooms (Virtuosi Hall Room V002 ~ V009) / Conference Hall
10:30~11:00	Coffee & Tea Break	
11:00~11:10	Opening Ceremony	Virtuosi Hall Room V001
11:10~12:00	Keynote Speech I (Prof. Yiran Chen, Duke University, USA)	Virtuosi Hall Room V001
12:00~13:00	Lunch	
13:10~14:00	Keynote Speech II (Prof. Tsu-Chin Tsao, The University of California, Los Angeles (UCLA), USA)	Virtuosi Hall Room V001
14:00~15:30	Sessions Track 2 / NSTC Poster 2 / Best Paper 1/ Best Student Paper 1 / Sponsor Spotlight	Session Rooms (Virtuosi Hall Room V002 ~ V009) / Conference Hall
15:30~15:50	Coffee & Tea Break	
15:50~17:20	Sessions Track 3 / Best Paper 2 / Best Student Paper 2	Session Rooms (Virtuosi Hall Room V002 ~ V009)
17:20~18:20	Free Shuttle Bus Departure	
18:00~20:00	Welcome Reception (Dinner)	InSky Hotel (2F Restaurant)

Friday, October 23, 2026

Time	Topic	Venue
08:00~17:00	Registration & Exhibition	Virtuosi Hall Conference Lobby
09:00~10:00	Keynote Speech III (Prof. Hai Li, Duke University, USA)	Virtuosi Hall Room V001
10:00~10:30	Coffee & Tea Break	
10:30~12:00	Sessions Track 4 / NSTC Poster 3	Session Rooms (Virtuosi Hall Room V001 ~ V009) / Conference Hall
12:00~13:00	Lunch	

13:00~14:30	Sessions Track 5	Session Rooms (Virtuosi Hall Room V001 ~ V009)
14:30~14:50	Coffee & Tea Break	
14:50~16:05	Sessions Track 6	Session Rooms (Virtuosi Hall Room V001 ~ V009)
16:05~16:40	Invited Lecture (Prof. Chun-Liang Lin)	Virtuosi Hall Room V001
16:40~17:30	國科會控制學門業務報告 (Prof. Chia-Feng Juang)	Virtuosi Hall Room V001
17:30~18:20	Free Shuttle Bus Departure	
18:20~20:30	Conference Banquet	Truelove Garden Hotel (Room 9)

Saturday, October 24, 2026

Time	Topic	Venue
Full Day	One Day Tour – Sun Moon Lake	Nantou, Taiwan

Program at a Glance

Date	Time	Location (Virtuosi Hall Conference Lobby) (共善樓)			
Oct. 22 *(Thu.)*	08:00 - 17:00	Registration & Exhibition			
		Room V001	Room V002	Room V003	Room V004
			R-1: Advanced Control Technologies and Applications	SS-02: Agentic and Physical AI-Driven Green Energy Transformation toward Net-Zero Energy Systems	SS-08: Computational Intelligence Methods and Explainable AI for Evaluation and Distillation of AI Data Sets
			Chair: Jau-Woei Perng	Chair: Chun-Wei Huang	Chair: Chen-Chia Chuang
	09:00 - 09:15		1025	1024	1090
	09:15 - 09:30		1048	1095	1117
	09:30 - 09:45		1052	1243	1138
	09:45 - 10:00		1280	1257	1158
	10:00 - 10:15			1270	1172
	10:15 - 10:30			1010	1241
	10:30 - 11:00	Coffee & Tea Break			
	11:00 - 11:10	Opening Ceremony			
	11:10 - 12:00	Keynote Speech I Prof. Yiran Chen (Duke University, USA)			
	12:00 - 13:00	Lunch / CAC Board Meeting (Room: V003)			
	13:10 - 14:00	Keynote Speech II Prof. Tsu-Chin Tsao (The University of California, Los Angeles (UCLA, USA)			
			BC-1: Best Conference Paper Contest	BS-1: Best Student Paper Contest	SS-05-1: Intelligent Mobile Robotic Systems and Applications
					Chair: Hsu-Chih Huang
	14:00 - 14:15	Sponsor Spotlight	1036	1026	1065
	14:15 - 14:30		1092	1032	1139
	14:30 - 14:45		1152	1144	1146
	14:45 - 15:00		1203	1176	1230
	15:00 - 15:15		1235	1190	1297
	15:15 - 15:30			1205	
	15:30 - 15:50	Coffee & Tea Break			
			BC-2: Best Conference Paper Contest	BS-2: Best Student Paper Contest	SS-05-2: Intelligent Mobile Robotic Systems and Applications
					Chair: Ching-Chih Tsai
	15:50 - 16:05		1091	1217	1003
	16:05 - 16:20		1099	1221	1004
	16:20 - 16:35		1108	1222	1005
	16:35 - 16:50		1110	1254	1123
	16:50 - 17:05		1290	1275	
	17:05 - 17:20				
	17:20 - 18:00	Transportation (Free Shuttle Bus)			
	18:00 - 20:00	Welcome Reception (Dinner)			

Date	Time	Location (Virtuosi Hall Conference Lobby) (共善樓)				
Oct. 22 *(Thu.)*	08:00 - 17:00	Registration & Exhibition				
		Room V005	Room V006	Room V007	Room V008	Room V009
		SS-10-1: Intelligent Control and Applications	SS-11: Artificial Intelligence-Based Preclinical and Clinical Research on Alzheimer's Disease, Epileptic EEG, and Noninvasive Brain Stimulation	SS-20: Multimodal AI, Computer Vision, and Intelligent Systems for Real-World Applications	SS-33-1: Smart Robotics and Intelligent Systems Applications	SS-35: Artificial Intelligence and Multi-Modality Imaging in Precision Radiation Therapy
		Chair: Gwo-Ruey Yu	Chair: Yue-Der Lin	Chair: Sze-Teng Liong	Chair: Chiu-Hung Chen	Chair: Weir-Chiang You
	09:00 - 09:15	1072	1127	1016	1049	1155
	09:15 - 09:30	1079	1207	1077	1066	1157
	09:30 - 09:45	1094	1225	1142	1081	1199
	09:45 - 10:00	1120	1249	1185	1122	1202
	10:00 - 10:15	1209	1296	1218	1177	
	10:15 - 10:30	1210		1009		
	10:30 - 11:00	Coffee & Tea Break				
	11:00 - 11:10	Opening Ceremony				
	11:10 - 12:00	Keynote Speech I Prof. Yiran Chen (Duke University, USA)				
	12:00 - 13:00	Lunch / CAC Board Meeting (Room: V003)				
	13:10 - 14:00	Keynote Speech II Prof. Tsu-Chin Tsao (The University of California, Los Angeles (UCLA, USA)				
		SS-10-2: Intelligent Control and Applications	SS-17-1: Data-Driven Control and Learning for Autonomous Mobile Systems	SS-34: Smart Home Service Robot	SS-33-2: Smart Robotics and Intelligent Systems Applications	SS-24-1: Integrated Monitoring and Control
		Chair: Chiang-Ju Chien	Chair: Chia-Wen Chang	Chair: Hsuan-Ming Feng, Chi-Yi Tsai	Chair: Yu-Kun Lin	Chair: Yue-Shan Chang
	14:00 - 14:15	1029	1050	1104	1042	1039
	14:15 - 14:30	1191	1061	1136	1219	1134
	14:30 - 14:45	1239	1064	1264	1220	1147
	14:45 - 15:00	1242	1098	1292	1238	1182
	15:00 - 15:15	1274	1256	1293	1277	
	15:15 - 15:30		1263			
	15:30 - 15:50	Coffee & Tea Break				
		SS-13: Evolutionary Computation and Deep Learning for Autonomous Perception and Decision-Making	SS-17-2: Data-Driven Control and Learning for Autonomous Mobile Systems	SS-14: System Integration and Control / SS-22: Robust Quantum Filter and Control Design and Their Applications to Quantum Processing and Communication	SS-15: Applications of Generative Artificial Intelligence on Robotics	SS-24-2: Integrated Monitoring and Control
		Chair: Jia Guo	Chair: Hsin-Han Chiang	Chair: Fu-Cheng Wang, Bor-Sen Chen	Chair: Ching-Kun Chen	Chair: Meng-Yun Chung
	15:50 - 16:05	1119	1162	1021	1013	1017
	16:05 - 16:20	1148	1169	1031	1063	1022
	16:20 - 16:35	1186	1227	1033	1070	1076
	16:35 - 16:50	1226	1228	1067	1111	1107
	16:50 - 17:05	1288	1262	1246		1260
	17:05 - 17:20		1285	1267		
	17:20 - 18:00	Transportation (Free Shuttle Bus)				
	18:00 - 20:00	Welcome Reception (Dinner)				

Date	Time	Location (Virtuosi Hall Conference Lobby) (共善樓)			
Oct. 23 *(Fri.)*	08:00 - 17:00	Registration & Exhibition			
		Room V001	Room V002	Room V003	Room V004
	9:00 - 10:00	Keynote Speech III Prof. Hai Li (Duke University, USA)			
	10:00 - 10:30	Coffee & Tea Break			
		SS-12-1: AI-Driven Perception, Prediction, and Decision-Making for Autonomous Systems	SS-01: Smart System and Its Applications	SS-06: Development of Intelligent Hearing Aids and Heart Rhythm Care Devices	SS-16: Computer Vision for Intelligent Robot Applications
		Chair: Ching-Huei Huang	Chair: Sin-Ye Jhong	Chair: Cheng-Yuan Chang	Chair: Feng-Chi Lee
	10:30 - 10:45	1014	1113	1084	1047
	10:45 - 11:00	1030	1118	1093	1140
	11:00 - 11:15	1105	1204	1154	1265
	11:15 - 11:30	1133	1212	1245	1276
	11:30 - 11:45	1137	1214		1278
	11:45 - 12:00	1183			
	12:00 - 13:00	Lunch / CAC General Assembly (Room: V003)			
		R-2: Advanced Control Technologies and Applications	SS-04: Data-driven Computational Intelligence Techniques and Applications	SS-09-1: AI Based Robotics	SS-23-1: Development and Integration of Intelligent Medical Diagnosis and Rehabilitation Care Systems
		Chair: Chan-Yun Yang	Chair: I-Fang Chung	Chair: Wen-June Wang	Chair: Chih-Min Lin
	13:00 - 13:15	1071	1023	1073	1075
	13:15 - 13:30	1173	1097	1088	1125
	13:30 - 13:45	1196	1116	1184	1194
	13:45 - 14:00	1197	1166	1271	1195
	14:00 - 14:15	1261	1251	1287	1229
	14:15 - 14:30		1295		1231
	14:30 - 14:50	Coffee & Tea Break			
		SS-32: BrainSynergy: An Open-Source AI Platform for Multimodal Neuroimaging and Applications Across the Lifespan	SS-12-2: AI-Driven Perception, Prediction, and Decision-Making for Autonomous Systems	SS-09-2: AI Based Robotics	SS-23-2: Development and Integration of Intelligent Medical Diagnosis and Rehabilitation Care Systems
		Chair: Jeen-Shing Wang	Chair: Wei-Lun Lin	Chair: Wen-June Wang	Chair: Jing-Chen Hong
	14:50 - 15:05	1135	1151	1085	1074
	15:05 - 15:20	1188	1159	1129	1115
	15:20 - 15:35	1234	1192	1132	1128
	15:35 - 15:50	1248	1266	1181	1171
	15:50 - 16:05	1283	1294	1284	
	16:05 - 17:30	國科會控制學門業務報告 Prof. Chia-Feng Juang			
	17:30 - 18:20	Transportation (Free Shuttle Bus)			
	18:20 - 20:30	Conference Banquet			

Date	Time	Location (Virtuosi Hall Conference Lobby) (共善樓)				
Oct. 23 *(Fri.)*	08:00 - 17:00	Registration & Exhibition				
		Room V005	Room V006	Room V007	Room V008	Room V009
	9:00 - 10:00	Keynote Speech III Prof. Hai Li (Duke University, USA)				
	10:00 - 10:30	Coffee & Tea Break				
		SS-18-1: AI-Empowered Smart Driving Technology	SS-19-1: Intelligent Control in Robotics and Energy Applications	SS-21: Robotic Manipulator Technology and Applications	SS-26: Advanced Guidance, Navigation and Control Systems for Unmanned Aircraft Systems	SS-29: Observer-Based Control Design for Mechatronic Systems
		Chair: Yen-Lin Chen	Chair: Sei Zhen Khong	Chair: Syuan-Yi Chen	Chair: Ying-Chih Lai	Chair: Ming-Yang Cheng
	10:30 - 10:45	1087	1015	1053	1143	1114
	10:45 - 11:00	1232	1040	1149	1170	1130
	11:00 - 11:15	1233	1045	1164	1198	1145
	11:15 - 11:30	1258	1020	1175	1253	1161
	11:30 - 11:45			1273	1268	
	11:45 - 12:00					
	12:00 - 13:00	Lunch / CAC General Assembly (Room: V003)				
		SS-18-2: AI-Empowered Smart Driving Technology	SS-19-2: Intelligent Control in Robotics and Energy Applications	SS-25: Adaptive and Learning Control of Intelligent Robotic Systems	SS-27: Real-Time Intelligent Systems and Control for Advanced Mechatronics	SS-31-1: Advanced Control and Analysis for Intelligent Mechatronic Systems
		Chair: Shih-Hsien Hsu	Chair: Kai-Yi Wong	Chair: Ming-Li Chiang	Chair: Nien-Tsu Hu	Chair: Shyh-Leh Chen
	13:00 - 13:15	1038	1059	1037	1054	1078
	13:15 - 13:30	1109	1101	1211	1102	1112
	13:30 - 13:45	1187	1215	1236	1106	1163
	13:45 - 14:00	1189	1269	1250	1126	1178
	14:00 - 14:15	1259	1286	1289	1206	1179
	14:15 - 14:30					1180
	14:30 - 14:50	Coffee & Tea Break				
		SS-03: Next-Generation Robotic Perception: Integrating Deep Learning and Machine Vision	SS-07: Automatic Control for Human-Robot Collaboration	SS-28: Designs and Applications of GAI based Service Robots	SS-30: Perception and Decision-Making of Intelligent Systems	SS-31-2: Advanced Control and Analysis for Intelligent Mechatronic Systems
		Chair: Yi-Zeng Hsieh	Chair: Po-Ting Lin	Chair: Tzuu-Hseng S. Li	Chair: Chen-Chien Hsu	Chair: Woraphrut Kornmaneesang
	14:50 - 15:05	1028	1012	1034	1069	1051
	15:05 - 15:20	1174	1060	1035	1131	1103
	15:20 - 15:35	1216	1062	1043	1165	1208
	15:35 - 15:50	1224	1121	1080	1168	1237
	15:50 - 16:05	1240	1213	1255	1223	1272
	16:05 - 17:30	國科會控制學門業務報告 Prof. Chia-Feng Juang				
	17:30 - 18:20	Transportation (Free Shuttle Bus)				
	18:20 - 20:30	Conference Banquet				

CACS Technical Session Details

Session Track 1

Oct. 22, 2026, 09:00 ~ 10:30

CACS-R-01 **(Room V002)** Thursday, Oct. 22, 2026, 09:00 ~ 10:30

TOPIC: ADVANCED CONTROL TECHNOLOGIES AND APPLICATIONS

Session Chair : Prof. Jau-Woei Perng

Time	Paper ID	Title & Author
09:00-09:15	**#1025**	**Multi-Objective Optimization of Sampling Frequency and Window Configurations for Fall Detection on the SisFall Dataset** I-An Lin, Li-Cheng Zhong, Sung-Yun Hsiao, Tian-Feng Wang
09:15-09:30	**#1048**	**A Low-Cost Portable A-Mode Ultrasound System for ToF-Based Muscle Thickness Estimation** Tien-Yu Chang, ying Huang, Ben-Yi Liau
09:30-09:45	**#1052**	**Development of a Human Digital Twin for Biomechanical Gait Analysis and Deep Reinforcement Learning-Based Balance Control** Ye-Cheng Zeng, Liang-Pin Chou, You-Han Chen, Chia-Ming Chang, Anh Quan Le Tran, Jia-Yush Yen, Wei-Li Hsu, Fu-Cheng Wang, Shun-Feng Su, Tsu-Chin Tsao
09:45-10:00	**#1280**	**Chirikov-Taylor Map Image Encryption System Based on DE1-SoC Heterogeneous Computing Architecture** Yu-Chieh Cheng, De-Sheng Chen

CACS-SS-02 **(Room V003)** Thursday, Oct. 22, 2026, 09:00 ~ 10:30

TOPIC: AGENTIC AND PHYSICAL AI-DRIVEN GREEN ENERGY TRANSFORMATION TOWARD NET-ZERO ENERGY SYSTEMS

Session Chair : Prof. Chun-Wei Huang

Time	Paper ID	Title & Author
09:00-09:15	**#1024**	**Short-Term Educational Building Power Prediction and Network Power Estimation Using Public Energy Data** Yun-Hsiang Chang, You-Jun Lin
09:15-09:30	**#1095**	**Design of DRL-based LLCL Filters Optimization for Grid-Forming Inverters** Chun Wei Huang, Ching An Chen, Wen Tso Wang
09:30-09:45	**#1243**	**Intelligent Speed Oscillation Suppression for PMSM Servo Drive Under Repetitive Load Disturbances** Shih-Gang Chen, Sin-Ruei Lee, Bo-Sheng Yang
09:45-10:00	**#1257**	**A Study on Improving Electric Vehicle Localization and Mapping Accuracy Based on Multi-Feature Fusion Loop Closure Detection** Shih-Hsien Hsu, Ju Chao, Min-Chieh Lin
10:00-10:15	**#1270**	**Regime-Specific Defensive Dispatch for Industrial Pseudo-NILM with Explicit Abstention** Chung Kuang Chen, Shu-Hao Liang
10:15-10:30	**#1010**	**Towards Green Manufacturing: A Cyber-Physical System Architecture for Net-Zero Energy Transformation and Carbon Footprint Integration in Precision Machinery** Chih-Yin Lin, Jiunn Fang, Kuo-Cheng Tai

CACS-SS-08 | **(Room V004)** Thursday, Oct. 22, 2026, 09:00 ~ 10:30

TOPIC: COMPUTATIONAL INTELLIGENCE METHODS AND EXPLAINABLE AI FOR EVALUATION AND DISTILLATION OF AI DATA SETS

Session Chair : Prof. Chen-Chia Chuang

Time	Paper ID	Title & Author
09:00-09:15	**#1090**	**Early Yield Risk Prediction in Semiconductor Manufacturing Using WAT Data and PSO-Optimized Random Forest** Yuan-Jui Hsu, Ching-Hung Lee
09:15-09:30	**#1117**	**Reinforcement Learning-Based Hyperparameter Optimization for Speaker Diarization in Meeting Assistant Systems** Ti-Han Chuang, Hui-Ju Chen, Chen Chien Hsu, Chao-Hsiang Hsiao, Yin-Tien Wang
09:30-09:45	**#1138**	**An Intelligent Data Analysis Approach for Interval-Valued Data Based on a Bivariate Center-Range Regression Model** Chen-Chia Chuang
09:45-10:00	**#1158**	**Carbon Content Classification in Grayscale Spark Images and Confidence-Based Data Filtering Study** Shun-Feng Su
10:00-10:15	**#1172**	**Kernel-Based Graph Attention Networks with Broad Learning System Readout: Trading Expressiveness for Closed-Form Incremental Learning** Yan-Hua Su, Jin-Tsong Jeng
10:15-10:30	**#1241**	**Hyperparameter Optimization of an ANN-Based Synchronization Error Model for CNC Rigid Tapping Using Simulated Annealing** Sung-Yu Tsai, Po-Yuan Yang, Fu-I Chou

CACS-SS-10-01	(Room V005) Thursday, Oct. 22, 2026, 09:00 ~ 10:30	
TOPIC: INTELLIGENT CONTROL AND APPLICATIONS		
Session Chair : Prof. Gwo-Ruey Yu		
Time	**Paper ID**	**Title & Author**
09:00-09:15	**#1072**	**A Multi-Strategy Enhanced Hybrid Artificial Protozoa Optimizer for High-Dimensional Global Optimization** Yu-Ting Su, Kuo-Hsiang Yan, Wei-Hung Cheng
09:15-09:30	**#1079**	**A Two-Stage Reinforcement Learning Framework with Gated Rewards for Dexterous Insertion** Chun-Yu Hsu, Yu-Ping Liao, Chian-Song Chiu
09:30-09:45	**#1094**	**IoT and Machine-Learning-Based Water Quality Risk Prediction and Control for Aquaculture** Yong-Du Huang, Zhe-Kai Chiang, Jhe-Wei Lin, Chun-Hsiu Yeh, Hsin-Hua Chen
09:45-10:00	**#1120**	**Passivity-Based Interval Type-2 Fuzzy Covariance Control for Ship Fin Stabilizers under Multiplicative Noises and Input-Channel Deception Attacks** Yann-Horng Lin, Wen-Jer Chang, En-Chi Kuo
10:00-10:15	**#1209**	**Photovoltaic Panel Defect Inspection Based on Quantum-Inspired QSH-YOLO26s** Z.-Q. Wang, C.-D. Su, Gwo-Ruey Yu,
10:15-10:30	**#1210**	**Quantum Deep Learning-Based Visual SLAM System for Quadrotors** S.-H. Yu, Jyun-Ting Chen, Gwo-Ruey Yu

CACS-SS-11 **(Room V006)** Thursday, Oct. 22, 2026, 09:00 ~ 10:30

TOPIC: ARTIFICIAL INTELLIGENCE-BASED PRECLINICAL AND CLINICAL RESEARCH ON ALZHEIMER'S DISEASE, EPILEPTIC EEG, AND NONINVASIVE BRAIN STIMULATION

Session Chair : Prof. Yue-Der Lin

Time	Paper ID	Title & Author
09:00-09:15	**#1127**	**EEGAN-MR: Robust EEG Feature Augmentation for Early Cognitive Impairment Detection** Yi-Hung Liu, Hu Hong-Yuan, Po-Chien Chi, Wei-Ting Hung
09:15-09:30	**#1207**	**Design of an electrical stimulation system based on inferences about hand movements using high-frequency modulated current and array electrodes** Hong-Yan Liao, Po-Lei Lee
09:30-09:45	**#1225**	**Source-Level EEG Functional Connectivity and Graph Attention Networks for Cognitive Function Assessment** Mariya Nissar, Po-Lei Lee
09:45-10:00	**#1149**	**Residual Reinforcement Learning with Frozen Behavior-Cloning Policy for Robust Sim-to-Real Robotic Pick-and-Place** Hong-Si Su, Rui-Hua Xu, Guan-An Chen, Syuan-Yi Chen
10:00-10:15	**#1296**	**CONFER: Connectivity-Aware EEG Reconstruction from a Fixed 8-Channel Montage with a Band-Limited Flow-Matching Residual** Tung-Lun Yang, Kuo-Kai Shyu, Po-Lei Lee, Szu-Chi Huang

CACS-SS-20 **(Room V007)** Thursday, Oct. 22, 2026, 09:00 ~ 10:30

TOPIC: MULTIMODAL AI, COMPUTER VISION, AND INTELLIGENT SYSTEMS FOR REAL-WORLD APPLICATIONS

Session Chair : Prof. Sze-Teng Liong

Time	Paper ID	Title & Author
09:00-09:15	**#1016**	**A Lightweight Browser-Based Volleyball Motion Analysis System Using 3D Pose and Hand Landmark Features** Yu Wei Chiu, Pei Ling Chen, Ting Yu Lin, Yun Ching Lee, Heng-Fu Zhang, Ben-Yi Liau
09:15-09:30	**#1077**	**Automatic Pig Shipment Counting Using YOLO-Based Detection and Cross-Zone Center-Point Analysis** Yu-Ming Huang, Heng-Fu Zhang, Ben-Yi Liau
09:30-09:45	**#1142**	**An Automated Multi-Class Lumbar Spine Segmentation System Based on U-Net for Frontal X-Ray Images** Sze Teng Liong, Chi-Hsuan Lin, Wei-Chun Lu, Yen-Wei Tsao, Min-Zhen Wu, TsungYi Chen, Tzong-Rong Ger, Shih-Lun Chen, Tsun-Kuang Chi, Patricia Angela R. Abu
09:45-10:00	**#1185**	**A Dual-Branch Framework for Semantic Segmentation and Small Object Detection of Surface Floating Waste** Hsi-Hung Wang, Yao-Cheng Wu, Kuan-Che Huang, Yu-Shiuan Tsai
10:00-10:15	**#1218**	**Automatic Curve-Based Root Canal Filling Length Measurement on Periapical Radiographs Using Tooth Segmentation and Filling Trajectory Reconstruction** Heng-Chia Chen, Yung-Hsin Liu, Po-Jui So, Kai-Syun Chiu, Wei-Chun Lu, Li Wei Ou-Yang, Tsun-Kuang Chi, TsungYi Chen, Tzong-Rong Ger, Patricia Angela R. Abu

10:15-10:30	**#1009**	**An AI Pedestrian Guardian Road Risk Warning System Based on YOLO26 and GPS-Based Environmental Awareness** Ms. Pei-Shan Peng, Ms. Chen-Wei Su, Ms. Sheng-Hsun Su, Ms. Heng-Fu Zhang, Ms. Ben-Yi Liau

CACS-SS-33-01 **(Room V008)** Thursday, Oct. 22, 2026, 09:00 ~ 10:30

TOPIC: SMART ROBOTICS AND INTELLIGENT SYSTEMS APPLICATIONS

Session Chair : Prof. Chiu-Hung Chen

Time	Paper ID	Title & Author
09:00-09:15	**#1049**	**Hydrophone and Wavelet Packet Transform Analysis for Monitoring a Diamond Wheel during Silicon Carbide Wafer Grinding** Ting-Yi Lee, Yu-Kun Lin
09:15-09:30	**#1066**	**Multi-Channel Speech Reverberation Removal Based on Diffusion Model: An Indoor Acoustics Case Study** Chiu-Hung Chen, Yen-Chi Li, Hao-Hsiang Hsu
09:30-09:45	**#1081**	**E-ROI-Based Thermal Dangerous-Object Recognition for Intelligent Inspection Robots** Chien-Jung Huang, Ying-Hsuan Liao, Yu-Kun Lin
09:45-10:00	**#1122**	**FPGA-Based Adaptive Hierarchical Median Estimation for Robust RGB-D Object Ranging** Chuan-Tsai Lin, Ting-Sheng Lin
10:00-10:15	**#1177**	**Design and Experimental Validation of a UWB-Assisted Coarse-Pointing System for UAV Optical Communication Ground Stations** Yu-Hung Wang, You-Yu Hsu, Li-An Tsai, Jhen-Syuan Huang

CACS-SS-35 **(Room V009)** Thursday, Oct. 22, 2026, 09:00 ~ 10:30

TOPIC: ARTIFICIAL INTELLIGENCE AND MULTI-MODALITY IMAGING IN PRECISION RADIATION THERAPY

Session Chair : Prof. Weir-Chiang You

Time	Paper ID	Title & Author
09:00-09:15	**#1155**	**Application of Automated Multi-Modality Images in Radiation Therapy for Prostate Cancers** Chia-Yu Lai, Tzuhsuan Chen, Weir-Chiang You, Chun Yi Lin, Chi-Wen Jao, Chia-Chi Wen, Chung-Hsien Hsu, Mau-Shin Chi, Yu-Te Wu, Kai-Lin Yang
09:15-09:30	**#1157**	**Federated Learning for Clinical Target Volume Segmentation in Cervical Cancer: A Two-Center Feasibility Study** Chun Yi Lin, Chia-Yu Lai, Wei-Kai Lee, Kai-Lin Yang, Kai-Hsuan Yao, Chi-Wen Jao, Ying-Yi Hsu, Yung-Fa Lu, Weir-Chiang You, Yu-Te Wu
09:30-09:45	**#1199**	**MedGemma-Guided Segmentation of Intact and Postoperative Cervical Cancer Target Volumes** Chia-Yu Lai, Kai-Lin Yang, Chun Yi Lin, Kai-Hsuan Yao, Chi-Wen Jao, Ying-Yi Hsu, Yung-Fa Lu, Yu-Te Wu, Weir-Chiang You
09:45-10:00	**#1202**	**Automated Seminal Vesicle Invasion Classification Using MRI Radiomics and nnU-Net-Based Segmentation** Tzuhsuan Chen, Chia-Yu Lai, Weir-Chiang You, Chun Yi Lin, Chi-Wen Jao, Chia-Chi Wen, Chung-Hsien Hsu, Mau-Shin Chi, Yu-Te Wu, Kai-Lin Yang

Session Track 2

Oct. 22, 2026, 14:00 ~ 15:30

CACS-BC-01 **(Room V002)** Thursday, Oct. 22, 2026, 14:00 ~ 15:30

TOPIC: BEST CONFERENCE PAPER CONTEST

Session Chair :

Time	Paper ID	Title & Author
14:00-14:15	**#1036**	**Design, Implementation, and Reinforcement Learning-Based Control Design for TaiwanBot Junior** Chia-Cheng Hsin, Shi-Han Wen, Chieh-Lun Tsai, Dong-Hua Wu, Ching-Yao Fang, Po-Yi Liu, Hou-Shin Wu, Tzuu-Hseng S. Li
14:15-14:30	**#1092**	**On Data-Driven Control-Theoretic Kelly Trading** Yu-Han Syu, Chung-Han Hsieh
14:30-14:45	**#1152**	**Intelligent Two-in-One Motion Control for Autonomous Medical Delivery Robots Using Fuzzy Wavelet Neural Networks with Output Recurrent Broad Learning Systems** Ching-Chih Tsai, Hao-Tang Wang, Zhong-Kai Wu
14:45-15:00	**#1203**	**Improved Dynamic Flux-Weakening Control for SynRM via Phase-Lead Compensation of Voltage Angle** Shih-Gang Chen, Sin-Ruei Lee
15:00-15:15	**#1235**	**Robot Imitation Learning Using Data Fusion of Images and Point Clouds for Cluttered Grasping and Cross-Scenario Manipulation** Kai-Tai Song, Yen Chih-Hsien

CACS-BS-01 **(Room V003)** Thursday, Oct. 22, 2026, 14:00 ~ 15:30

TOPIC: BEST STUDENT PAPER CONTEST

Session Chair :

Time	Paper ID	Title & Author
14:00-14:15	**#1026**	**A Synergistic Algorithmic Approach to Detect Chess Pieces for Autonomous SCARA Robot** Swarnajit Bhattacharya, Hsien-I Lin
14:15-14:30	**#1032**	**Development of an EDF-Based Wall-Climbing Robot for Multi- Mode Locomotion on Realistic Building Facades** Chia-Hung Kuo, Chi-Ying Lin
14:30-14:45	**#1144**	**A Scenario-Driven Vision-Language Decision Framework for End-to-End Autonomous Driving** Jhan-Yu Liao, Bo-Wei Wang, Yu-Chen Lin, Yu-Rui Chen, Cheng-Hsien Wang
14:45-15:00	**#1176**	**Robust 3D Object Detection for Solid-State LiDAR Perception on Autonomous Mobile Platforms** Yen-Yu Hsieh, Xu-Xiang Weng, Po-Shao Liu, Yi-Heng Sun, Yung-Yao Chen, Sin-Ye Jhong
15:00-15:15	**#1190**	**Query-Guided Support Rectification for Few-Shot Fine-Grained Image Classification** Derui Zheng, Jiacheng Li, Masato Noto
15:00-15:30	**#1205**	**An Omnidirectional Rehabilitation System Integrating Adaptive Physics-Informed Neural Network-Based Inverse Dynamics Control and Mixed-Reality Guidance** Hsin-Yi Lin, Shih-Wei-Chen Tseng, Lian-Wang Lee

CACS-SS-05-01 **(Room V004)** Thursday, Oct. 22, 2026, 14:00 ~ 15:30

TOPIC: INTELLIGENT MOBILE ROBOTIC SYSTEMS AND APPLICATIONS

Session Chair : Prof. Hsu-Chih Huang

Time	Paper ID	Title & Author
14:00-14:15	**#1065**	**Design of an Environmental Resource Exploration Rover Integrating an Autonomous Mobile Platform and a Robotic Arm** Ming-Chiuan Shiu, Tsung-Chi Tsai
14:15-14:30	**#1139**	**A GWO-HMM Framework for Lane Departure and Lane Change Recognition Based on Temporal Lateral Motion Patterns** Yun-Ling Chang, Hsin-Tung Ma
14:30-14:45	**#1146**	**ROS 2-Based Autonomous Mobile Power Delivery Robot Using Multimodal Sensor Fusion and Dynamic Energy Management** Ke-Jie Huang, Yu-Heng Chou, Shin-Hung Chang, Yung-Chun Sun, Han-Sheng Lu
14:45-15:00	**#1230**	**A Multi-Sensor Risk-Aware Navigation and Campus Tour Guidance System for an Intelligent Wheelchair** Jun-Wei Pai, Han-Pang Huang
15:00-15:15	**#1297**	**High efficiency particle collection in digital microfluidics on coplanar electrodes** Sung Lin TSAI, Chang Tzu-Ping, Shih-Hung Lin, Chen Chun-Hong

CACS-SS-10-02 **(Room V005)** Thursday, Oct. 22, 2026, 14:00 ~ 15:30

TOPIC: INTELLIGENT CONTROL AND APPLICATIONS

Session Chair : Prof. Chiang-Ju Chien

Time	Paper ID	Title & Author
14:00-14:15	**#1029**	**SOS ET Control of Ship Autopilot Servo System with Time-Varying Parameter** Cheung-Chieh Ku, Shou-Chih Huang
14:15-14:30	**#1191**	**Performance and Robustness of a New Sampled-data PD-type Iterative Learning Controller** Chiang-Ju Chien, Thuy Thi-Nhu Ngo
14:30-14:45	**#1239**	**Forecast Driven Communication Scheduling for Koopman Based Multirobot Formation Control** Md Hasibuzzaman, Chan-Yun Yang, Gene Eu Jan
14:45-15:00	**#1242**	**Physics-Informed Neural Network-Based Inverse Dynamics Control of a Pneumatic Parallel Manipulator** Guan-Yu Lin, u-Han Huang, Lian-Wang Lee
15:00-15:15	**#1274**	**Democracy-based Task Allocation in a Large-scale Multi-robot System** Ching-Wei Chuang, Hsien-Te Fu

CACS-SS-17-01 **(Room V006)** Thursday, Oct. 22, 2026, 14:00 ~ 15:30

TOPIC: DATA-DRIVEN CONTROL AND LEARNING FOR AUTONOMOUS MOBILE SYSTEMS

Session Chair : Prof. Chia-Wen Chang

Time	Paper ID	Title & Author
14:00-14:15	**#1050**	**Remarks on Hypercomplex-Valued Deep Echo State Networks with Application to Trajectory-Tracking Control of a Quadrotor UAV** Kazuhiko Takahashi, Ibuki Kikuchi, Takuma Toyoda, Shinga Yoshitake, Masafumi Hashimoto
14:15-14:30	**#1061**	**Integration of Reinforcement Learning with LQR for Optimized Quadcopter Control** Min-Fan Ricky Lee, Ida Bagus Kade Rainata Putra Wibawa, Ingwer Ludwig Nommensen, Tzu-Chien Joey Tseng
14:30-14:45	**#1064**	**LSTM-Based Adaptive Measurement-Noise Covariance Estimation for Tightly Coupled UWB/IMU ESKF Localization** Yun Chiao Lai, Yu-Hsin Hsu, Jyh-Ching Juang
14:45-15:00	**#1098**	**A Vision-Guided Bowling Robot with Adaptive Strategy Selection** Chun-Fei Hsu, Chun-Hao Liao
15:00-15:15	**#1256**	**Deep State Space Model Predictive Control for Quadrotor Trajectory Tracking** KyungSoo Kim, Minseon Gwak, Hae Seong Lee, Poogyeon Park
15:00-15:30	**#1263**	**Sensorless Torque Estimation in Pushrim-Activated Power-Assisted Wheelchairs via a Four-State Constant-Jerk Observer** Chun-Lung Cheng, Yuan-Ting Lin, Jun-Zai Lin

CACS-SS-34	**(Room V007)** Thursday, Oct. 22, 2026, 14:00 ~ 15:30	
TOPIC: SMART HOME SERVICE ROBOT		
Session Chair : Prof. Hsuan-Ming Feng		
Time	**Paper ID**	**Title & Author**
14:00-14:15	**#1104**	**Vision-Only LoRA Adaptation of VLA Models with Limited Real-World Data** Huai-Tsung Yeh, Chi-yi Tsai
14:15-14:30	**#1136**	**Semantic-Driven Multi-Task Scheduling Mobile Robot Navigation System** Hsuan-Ming Feng, Ching-Chang Wong, Shih-An Li, Yun-Hui Chu
14:30-14:45	**#1264**	**Task Reasoning for Home Service Robots Using Large Language Models and Semantic Maps** Kun-Duo Weng, HSIEN-MIN TAI, hung-yu Liao, Chao-Yun Hsu, Yi-Chan Wu, Hung-Hsuan Cheng, Hsuan-Ming Feng, Ching-Chang Wong
14:45-15:00	**#1292**	**Design of a Wheeled Mobile Dual-Arm Robot for Home Service** Ya-Wen Liu, Kun-Duo Weng, HSIEN-MIN TAI, Hung-Yu Liao, Wei-Che Liao, Hsuan-Ming Feng, Ching-Chang Wong
15:00-15:15	**#1293**	**Vision-Language Model-Based Semantic-Augmented Map Construction** Kun-Duo Weng, Hsien-Min Tai, Hung-Yu Liao, Tzu-Yi Huang, Hsuan-Ming Feng, Ching-Chang Wong

CACS-SS-33-02 **(Room V008)** Thursday, Oct. 22, 2026, 14:00 ~ 15:30

TOPIC: SMART ROBOTICS AND INTELLIGENT SYSTEMS APPLICATIONS

Session Chair : Prof. Yu-Kun Lin

Time	Paper ID	Title & Author
14:00-14:15	**#1042**	**Precision Spraying and Weeding via Integrated Navigation and Visual Servo Control on an Agricultural Vehicle** Po-Hua Lee, Shiang-Bin Wang, Su-Pin Lin, Pin-Yung Chen
14:15-14:30	**#1219**	**Smart Lightweight AI Dispensing Platform Based on Vision Verification and Pneumatic Sorting** Ching-huei Huang, Yen-Ching Lai, Yu-Chun Chou, Yu-Cheng Hsiao
14:30-14:45	**#1220**	**The Intelligent Spectrum Monitoring Platform for Passive RF Sensing in Public Safety** Ching-huei Huang, Pei-Zhen Jiang, Yen-Sheng Lee, Chen-Yu Li, Jui-Yang Chen
14:45-15:00	**#1238**	**Automated Wiring Inspection for Electrical Control Cabinets Using Artificial Intelligence-Based Image Recognition** Hsuan Li, Ping-Cheng Hsieh, Chi-Hsiang Ho
15:00-15:15	**#1277**	**Autoencoder-Based Health Indicator for Variable-Speed Gearbox Condition Monitoring: A Comparative Study of Statistical and Envelope/CWT Features** Kun-Ching Wang

CACS-SS-24-01 **(Room V009)** Thursday, Oct. 22, 2026, 14:00 ~ 15:30

TOPIC: INTEGRATED MONITORING AND CONTROL

Session Chair : Prof. Yue-Shan Chang

Time	Paper ID	Title & Author
14:00-14:15	**#1039**	**Intelligent MCU-based Lithium-ion Battery Preemptive Cooling Control for Remaining Driving Range Extension** Haohan Lee, Cong-Sheng Huang
14:15-14:30	**#1134**	**A Hybrid Model for Ocean Subsurface Temperature Prediction** Yue-Shan Chang, Bo-Xun Huang, Jiann-Horng Chen
14:30-14:45	**#1147**	**Design and Implementation of an ESP32-Based Three-Level Unidi-rectional Power Converter for Low-THD Applications** Meng-Yun Chung, Cheng-Chuan Chen, Xin-Ning Zhang, Pao-Hsuan Weng, Heng-Jui Chuang, Colaco Lanwin
14:45-15:00	**#1182**	**Development of a Human-Machine Collaboration System Using an Interacting Multiple Model** Chih-Jer Lin, Kun-Lin Tsai

Session Track 3

Oct. 22, 2026, 15:50 ~ 17:20

CACS-BC-02 **(Room V002)** Thursday, Oct. 22, 2026, 15:50 ~ 17:20

TOPIC: BEST CONFERENCE PAPER CONTEST

Session Chair :

Time	Paper ID	Title & Author
15:50-16:05	**#1091**	**Cluster-Based Adaptive Image Captioning for Enhancing Visual Content Descriptions in Multimodal RAG** Yen-Hua Lu, Ching-Hung Lee
16:05-16:20	**#1099**	**Design and Experimental Validation of a Dual-Axis Force-Feedback System with LoRa-Based Remote Parameter Configuration** Ming-Yen Wei, Yan-Bin Jiang
16:20-16:35	**#1108**	**Design of a 3-Axis Articulated Manipulator Using Second-Order Sliding Mode Momentum Observer and RL-Based Robust Control Strategy** Ke-En Liao, Po-Yen Su, Jin-Liang Lin, Cheng-Ting Ho, Regina Florensia Hidayat, Jim-Wei Wu
16:35-16:50	**#1110**	**Effective Supervisory Control of Reinforcement Learning with Optimality Preservation** Peter Chen
16:50-17:05	**#1290**	**Wearable EEG Brain-Controlled Lightweight Lower-Limb Exoskeleton System for Stroke Rehabilitation** Ying-Hsu Huang, Shi-Yong Luo, Cong-Ying He, Po-Hsun Cheng, Li-Wei Ko

CACS-BS-02 **(Room V003)** Thursday, Oct. 22, 2026, 15:50 ~ 17:20

TOPIC: BEST STUDENT PAPER CONTEST

Session Chair :

Time	Paper ID	Title & Author
15:50-16:05	**#1217**	**Control Barrier Function Design via Backstepping for Safe Trajectory Tracking of a Mobile Robot** Fumiya Iwazaki, Yasuyuki Satoh
16:05-16:20	**#1221**	**Boundary-Cell Evaluation of Geometric Reshaping for Underwater Multi-Robot Formation Control** Takashi Arai, Shiho Oshiro
16:20-16:35	**#1222**	**Electro-Thermal Feature Fusion CNN for Inverter Fault Diagnosis in PMSM Drives** Po-Cheng Hung, Shin-Hung Chang
16:35-16:50	**#1254**	**An Integrated Learning Framework for Multi-Robot Coordination and Human–Robot Collaboration** Yi-Chieh Hsueh, Chong-En Wang, Zih-Ming Su, Wen-Huang Chen, Tzuu-Hseng S. Li
16:50-17:05	**#1275**	**Sim-to-Real Performance Analysis of a Custom Quadruped Robot Using Gait-Generator-Guided Reinforcement Learning** Heng-Hsuan Lee, Yi-Cheng Wang, Yu-Hsuan Chi, Yu-Mei Huang, Yu-Chung Kuo, and Chung-Hsien Kuo

CACS-SS-05-02 **(Room V004)** Thursday, Oct. 22, 2026, 15:50 ~ 17:20

TOPIC: INTELLIGENT MOBILE ROBOTIC SYSTEMS AND APPLICATIONS

Session Chair : Prof. Ching-Chih Tsai

Time	Paper ID	Title & Author
15:50-16:05	**#1003**	**Adaptive Dose-Aware Online Navigation for Indoor Rescue AMRs in Post-Nuclear-Disaster Environments Using RGB-D Depth Camera** Ching-Chih Tsai, Chung-Mou Chang, Ali Rospawan, Guo-Hua Yang
16:05-16:20	**#1004**	**Autonomous Navigation of Quadruped Robots in Rough and Uneven Environments Using Reinforcement Learning Control and YOLO** Zi-Rong Chiu , Ching-Chih Tsai
16:20-16:35	**#1005**	**Learning-Guided Online Informative Belief Planning for Moving Target Search under Partial Observability** Ching-Chih Tsai, Shu-Chih Chen, Yen-Jui Chia
16:35-16:50	**#1123**	**Dining-Mate: An Omnidirectional Visually Impaired-Friendly Restaurant Guiding Robot Based on Multi-Sensor Fusion and Safe Human-Robot Interaction** Kang Min Liu, Jiun Wei Chen, Yung Hung Hung, Chen Chih Cheng

CACS-SS-13 **(Room V005)** Thursday, Oct. 22, 2026, 15:50 ~ 17:20

TOPIC: EVOLUTIONARY COMPUTATION AND DEEP LEARNING FOR AUTONOMOUS PERCEPTION AND DECISION-MAKING

Session Chair : Prof. Jia Guo

Time	Paper ID	Title & Author
15:50-16:05	**#1119**	**Improving Fire-Scene Image-Text Matching via CLIP Fine-Tuning and Hard-Negative Training** Ziqi Chen, Jiacheng Li, Masato Noto
16:05-16:20	**#1148**	**Multi-Scale ECA Placement in YOLOv8s-Seg for Segmentation of Insect Damage on Cucumber Leaves** Yafu Lu, Jiacheng Li, Masato Noto
16:20-16:35	**#1186**	**Synthetic-to-Real Crowd State Classification Using Motion-Only Features** Seiichiro Yoshida, Jiacheng Li, Masato Noto
16:35-16:50	**#1226**	**An Explicit Vehicle-Reduction Intensification Framework for Lexicographic VRPSPDTW** Yang Zhang, Jiacheng Li, Jia Guo
16:50-17:05	**#1288**	**Cosine-Adaptive Retrospective Differential Evolution for Multi-Objective Robotic Grasp Configuration Optimization** Jia Guo, Kenta Urano, Takuro Yonezawa, Nobuo Kawaguchi

CACS-SS-17-02 **(Room V006)** Thursday, Oct. 22, 2026, 15:50 ~ 17:20

TOPIC: DATA-DRIVEN CONTROL AND LEARNING FOR AUTONOMOUS MOBILE SYSTEMS

Session Chair : Prof. Hsin-Han Chiang

Time	Paper ID	Title & Author
15:50-16:05	**#1162**	**System-Level Symmetric Winding Design for Common-Mode Noise Reduction in Inverter-Fed BLDC Fan Motors** Jin-Shyan Lee, Tzu-Liang Chiang
16:05-16:20	**#1169**	**Language-Guided Multi-Point Navigation Using DT-lite for Autonomous Mobile Robots** Hsin Han Chiang, Zhao-Yi Huang, Hsiu-Chi Chang
16:20-16:35	**#1227**	**Layered Traversability Mapping from 3D Point Clouds in Multi-Floor Indoor Environments** Lin Jian-Dong, Chia-Wen Chang, Chin-Wang Tao
16:35-16:50	**#1228**	**Safety-Oriented Continue-or-Abort Decision Making for Automated Lane Changes Using Confidence-Aware CNN-LSTM and Expert Reassessment** Li-Ta Lai, Chuan-Yu Yan, Hsin Han Chiang, Bo-Chiuan Chen
16:50-17:05	**#1262**	**Perceived-Safety-Enhanced Trajectory Planning for Automated Valet Parking via Improved Hybrid A* and Driver Risk Field** Hsin Han Chiang, Bing-Xiang Liu, Yue-Lin Tsai
17:05-17:20	**#1285**	**Optimization-Based Approach for Designing a Dynamic Output Feedback Controller for A Two-Wheeled Vehicle** Cheng-Han Shen, Jenq-Lang Wu

CACS-SS-14/22 **(Room V007)** Thursday, Oct. 22, 2026, 15:50 ~ 17:20

TOPIC: SYSTEM INTEGRATION AND CONTROL / ROBUST QUANTUM FILTER AND CONTROL DESIGN AND THEIR APPLICATIONS TO QUANTUM

Session Chair : Prof. Fu-Cheng Wang/Prof. Bor-Sen Chen

Time	Paper ID	Title & Author
15:50-16:05	**#1021**	**H-infinity-Motivated Risk-Aware TSP-QUBO Route Planning with Simulated Quantum Annealing** Jin-Kai Yeh, Chun-Liang Lin, Wei-Jhe Hsu
16:05-16:20	**#1031**	**Design and Modeling of a Controllable E-CVT Mechatronic Inerter Utilizing a Planetary Gear Set** I-Haur Tsai, Yi-Hsuan Lee, Sheng-Lei Wu
16:20-16:35	**#1033**	**Multiple Input Disturbance Decoupling Control for Mass Chains** Chun-Wei Liao, Tzu-Hsuan Tian, Fu-Cheng Wang
16:35-16:50	**#1067**	**Hybrid Position–Force Adaptive Control for Robot-Assisted Ultrasound Probe Manipulation** Sama Sun, Yu-Hsiu Lee
16:50-17:05	**#1246**	**Lyapunov-Based Bloch-Direction Tracking for a Two-Level Non-Markovian Open Quantum System** Cheng-Sin Shih, Ti-Chung Lee
17:05-17:20	**#1267**	**Quantum Communication System Design via Robust Quantum Reference Trajectory Tracking Control and Quantum Filter** Bor-Sen Chen, Shin-Yi Wen

CACS-SS-15 **(Room V008)** Thursday, Oct. 22, 2026, 15:50 ~ 17:20

TOPIC: APPLICATIONS OF GENERATIVE ARTIFICIAL INTELLIGENCE ON ROBOTICS

Session Chair : Prof. Ching-Kun Chen

Time	Paper ID	Title & Author
15:50-16:05	**#1013**	**Investigating the Capacities and Constraints of Small Language Models in the Context of Embodied Robotic Task Execution** Carolus Christadi Cahyono, Ivan Tanra, Hsien-I Lin, Louis Putra Purnama, Ade Septian Alfianto
16:05-16:20	**#1063**	**Lightweight LLM-Based Behavior Tree Generation with Dual-Layer Safety Validation for Fault-Tolerant Robot Pick-and-Place Tasks** He Hsiao, Kuu-Young Young
16:20-16:35	**#1070**	**Reliability-Aware LLM Task Planning for Industrial AMRs: Deterministic Validation and Cloud–Local Evaluation** Hao-Chun Pan, Kai-Hsien Ma, Hsuan-Hao Hsu, Chuan-Tsai Lin, Jau-Woei Perng
16:35-16:50	**#1111**	**MAG-BT: A Multimodal Adaptive Grounding Framework for LLM-Generated Behavior Trees in Contact-Rich Robotic Assembly** Shu-Chun Ho, Jhih-Yang Long, Kuu-Young Young

CACS-SS-24-02 **(Room V009)** Thursday, Oct. 22, 2026, 15:50 ~ 17:20

TOPIC: INTEGRATED MONITORING AND CONTROL

Session Chair : Prof. Meng-Yun Chung

Time	Paper ID	Title & Author
15:50-16:05	**#1017**	**Two-Stage Fan Blade Fault Diagnosis Using Three-Phase Motor Current-Derived Features** Jin-Yan Wu, Heng-Fu Zhang, Ben-Yi Liau
16:05-16:20	**#1022**	**Novel Battery Health Prognostics Based on Exponential Degradation Modeling: Applied to Intelligent Net Cage Aquaculture** Bo-Wei Huang, Yu-Hsiu Lin, Chun-Ting Su, Sheng-Ting Huang
16:20-16:35	**#1076**	**Hierarchical Multi-Task Deep Learning with Domain-Adaptive Spectral Matching for Heart Sound Diagnosis** Yi-Chieh Yu, Yun-Chia Chen, Yu-Te Wu, Chi-Wen Jao
16:35-16:50	**#1107**	**An Edge-AI Virtual Fence with Adaptive Sound Deterrence for Real-Time Monkey Intrusion Detection** Mochamad Rizal Fauzan, Irgi Surya, Yo-Ping Huang, Hwa-Ching Lin
16:50-17:05	**#1260**	**Relaxed admissibility criterion for delayed singular systems via extended Lyapunov-Krasovskii Functionals** Hae Seong Lee, KyungSoo Kim, Poogyeon Park

Session Track 4

Oct. 23, 2026, 10:30 ~ 12:00

CACS-SS-12-1 **(Room V001)** Friday, Oct. 23, 2026, 10:30 ~ 12:00

TOPIC: AI-DRIVEN PERCEPTION, PREDICTION, AND DECISION-MAKING FOR AUTONOMOUS SYSTEMS

Session Chair : Prof. Ching-Huei Huang

Time	Paper ID	Title & Author
10:30-10:45	**#1014**	**Edge-Deployed Risk Prediction for Taiwan-Specific Hazardous Driving Scenarios** Chih-Chung Hsu, Jia-Kai Zhong, Xin-Di Ma, Wen-Hai Tseng
10:45-11:00	**#1030**	**A Bounded-Authority Dual-Process Framework for Closed-Loop Vision-Language Autonomous Driving** Zhe-Zheng Shi, Wen Hui Chen, Yu-Chen Lin
11:00-11:15	**#1105**	**Threshold-Aware Semiconductor Fault Detection under Class Imbalance and Sensor-Data Missingness** Ting-Wei Wu, Kuan-Chu Lu
11:15-11:30	**#1133**	**From 2D Concept Segmentation to 3D Boxes: Training-Free Multi-Camera Auto-Labeling** Chia-Yu Chung, Chieh-Yu Tsai, Li-Wei Kang
11:30-11:45	**#1137**	**A Dynamic ROI and Dual-Axis Push-Broom Acoustic Array for Submarine Cable Inspection Using an Underwater Quadruped Robot** Jui Chen Yen, Tsai Sheng-Nan, Yu-Chen Lin, Hsu Chih-Yin, Yang Chen-En
11:45-12:00	**#1183**	**DQN-QoE Slice Admission Control for Live OAI Open RAN** Manoj Prasad Kunasegran, Wai Leong Pang, Swee King Phang

CACS-SS-01	**(Room V002)** Friday, Oct. 23, 2026, 10:30 ~ 12:00	
TOPIC: SMART SYSTEM AND ITS APPLICATIONS		
Session Chair : Prof. Sin-Ye Jhong		
Time	**Paper ID**	**Title & Author**
10:30-10:45	**#1113**	**CSPF: AChannel-Spatial Progressive Fusion Decoder for Simultaneous Medical Image Segmentation and Classification** Warawut Janwittayachai, Yulius Harjoseputro, Auzan Izzaturrahman, Phuc Xuan Duy Doan, Muhammad Zaki Mubarak Hariyadi, Albert William Wibisono, Yung-Yao Chen
10:45-11:00	**#1118**	**Development of intelligent single-phase energy metering devices based on embedded technology** Yun-Long Xie, Yu-Hsiu Lin, Chia-Hung Lai
11:00-11:15	**#1204**	**Temporal State-Space Detector for RGB-Event Object Detection in Driving Scenarios** Sin-Ye Jhong, Phuc Xuan Duy Doan, Yung-Yao Chen, Chao-Lung Yang
11:15-11:30	**#1212**	**A Hybrid Deep Learning and Rule-Based Geometric Measurement Architecture for Low-Cost Multi-View Multi-Task Inspection of Gas Valve Assemblies** Guan-Ming Lin, Sih-Hao Huang, Ding-En Dai, Chia-Hung Lai, Shu-Hsien Huang, Yu Chen Chien
11:30-11:45	**#1214**	**Intelligent Gear Abnormal Sound Detection Using STFT-Based Acoustic Features and Convolutional Neural Networks** Sih-Hao Huang, Chia-Hung Lai, Yu-Hsiu Lin, Shu-Hsien Huang

CACS-SS-06 **(Room V003)** Friday, Oct. 23, 2026, 10:30 ~ 12:00

TOPIC: DEVELOPMENT OF INTELLIGENT HEARING AIDS AND HEART RHYTHM CARE DEVICES

Session Chair : Prof. Cheng-Yuan Chang

Time	Paper ID	Title & Author
10:30-10:45	**#1084**	**Integration of smart sound amplification with noise reduction to headphone design** Cheng-Yuan Chang, Chong-Rui Huang, Jui-Yu Hsu
10:45-11:00	**#1093**	**An Edge-Intelligence Interactive System Based on Millimeter-Wave Radar and Environmental Sound Recognition** Yu-Ping Liao, Jia-Jung Li, Shih-Yun Huang, Li-Shan Ma
11:00-11:15	**#1154**	**Development and Performance Evaluation of an AI-Based Communication Aid for Individuals with Hearing Loss Using On-Device Speech Recognition and Speaker Diarization** Wan-Ju Tu, Yu-Chieh Wang, Li-Shan Ma, Shih-Hsiung Twu
11:15-11:30	**#1245**	**Hark: Evaluation of Speech Compensation Benefits in a Mobile Hearing Assistance System** Chia-Yin Wu, Cheng-Lun Tsai

CACS-SS-16 **(Room V004)** Friday, Oct. 23, 2026, 10:30 ~ 12:00

TOPIC: COMPUTER VISION FOR INTELLIGENT ROBOT APPLICATIONS

Session Chair : Prof. Ching-Kun Chen

Time	Paper ID	Title & Author
10:30-10:45	**#1047**	**QR-Nav: LLM-Based Mapless Zero-Shot Object Goal Navigation with Semantic Memory Queue and Re-perception Verification** Cheng Yu-Ting, Huei-Yung Lin
10:45-11:00	**#1140**	**High-Precision Positioning System for Badminton Courts Based on Active Vision and Depth Geometry Hybrid Correction at the Ball Post Perspective** Cheng-Hsien Wu, Hsuan-Kai Huang, Jeen-Shing Wang
11:00-11:15	**#1265**	**Optimization of Image Acquisition Strategies for WaterSplatting Reconstruction** Yung-Chi Liu, Pei Ju Chiang
11:15-11:30	**#1276**	**Vision-Guided Autonomous Manipulation of Humanoid Dual-Arm Robots via Imitation Learning** Ming-Yang Chuang, Han-Hao Liu, Chung-Hsien Kuo
11:30-11:45	**#1278**	**Cyber-Physical Systems Utilizing Guidance and Supervision Models for Automated Picking Tasks** Ting-Yu Chien, Shang-Chih Lin, Meng-Chia Wu, Shun-Feng Su, Yennun Huang

CACS-SS-18-01	**(Room V005)** Friday, Oct. 23, 2026, 10:30 ~ 12:00	
TOPIC: AI-EMPOWERED SMART DRIVING TECHNOLOGY		
Session Chair : Prof. Yen-Lin Chen		
Time	**Paper ID**	**Title & Author**
10:30-10:45	**#1087**	**Enhanced YOLO-based Recognition Deep-Learning Model using Lightweight CNN and Vision Transformer** Chao-Hsien Lee, Le-Rong Wu
10:45-11:00	**#1232**	**Training-Free Multi-Lane Detection From LiDAR Point Clouds via Intensity Clustering and Adaptive Node-Growth Tracking** Khean Thye Bea, Shiangde Tsai, Yen-Lin Chen
11:00-11:15	**#1233**	**Heterogeneous Sensor Data Collection and Spatiotemporal Calibration for a LiDAR–Camera Platform in Complex Taiwanese Traffic** Khean Thye Bea, Hsin-Yi Chen, Yen-Lin Chen
11:15-11:30	**#1258**	**Prioritized Experience Replay-Based Left-Turn Maneuver at Unsignalized Intersections for Self-Driving Cars** Hsin Han Chiang, Ze-Yu Fu, Yen-Lin Chen

CACS-SS-19-01 **(Room V006)** Friday, Oct. 23, 2026, 10:30 ~ 12:00

TOPIC: INTELLIGENT CONTROL IN ROBOTICS AND ENERGY APPLICATIONS

Session Chair : Prof. Sei Zhen Khong

Time	Paper ID	Title & Author
10:30-10:45	**#1015**	**Diffusion-Augmented Contrastive Bottleneck Network for Few-Shot Object Detection and Classification** Yu Jen Chen, Zih-Jyun Chen, Kao-Shing Hwang, Yu-Sheng Yang
10:45-11:00	**#1040**	**Foundation-Model Advantage at Scale: Data-Regime Crossovers in TCP-Deviation Prediction for Industrial Robot Arms** Cheng Chou Chang, Sei Zhen Khong
11:00-11:15	**#1045**	**PP-STP4: Probabilistic Time-Projected Spatiotemporal Path Planning for Robot Navigation in Dense Crowds** Ping Heng Hsieh, Kai-Yi Wong
11:15-11:30	**#1020**	**UAV-Based Intelligent Sentinel Surveillance System for Urban Target Monitoring** Min-Fan Ricky Lee, Tzu-Chien Joey Tseng, Jyun-Ruei Wu, Emirhan Sazakli, Yu-Chen Liu

CACS-SS-21 **(Room V007)** Friday, Oct. 23, 2026, 10:30 ~ 12:00

TOPIC: ROBOTIC MANIPULATOR TECHNOLOGY AND APPLICATIONS

Session Chair : Prof. Syuan-Yi Chen

Time	Paper ID	Title & Author
10:30-10:45	**#1053**	**Learning-Based Gain Adaptation and Command Shaping for Six-Degree-of-Freedom Manipulator Servo Control** Nien-Tsu Hu, Bo-Jhang Chen
10:45-11:00	**#1149**	**Residual Reinforcement Learning with Frozen Behavior-Cloning Policy for Robust Sim-to-Real Robotic Pick-and-Place** Hong-Si Su, Rui-Hua Xu, Guan-An Chen, Syuan-Yi Chen
11:00-11:15	**#1164**	**An LLM-Based Explainable Command Generation Model in Autonomous Driving System Design** Hsin-Hung Chen, Min-Jie Hsu
11:15-11:30	**#1175**	**Design and Development of an Adaptive Three-Finger Gripper with Multi-Sensor Integration for Robotic Manipulation** Li-Xuan Chen, Kai-Yang Tseng, Hsin-Cheng Lo, Yu-Rong Chen, Tsai-Yen Wei, Jeng-Dao Lee
11:30-11:45	**#1273**	**Hierarchical Next-Best-View Planning with Dense Semantic Mapping for Eye-in-Hand Robotic Manipulators** Yueh-Hsuan Lee, Han-Pang Huang

CACS-SS-26 **(Room V008)** Friday, Oct. 23, 2026, 10:30 ~ 12:00

TOPIC: ADVANCED GUIDANCE, NAVIGATION AND CONTROL SYSTEMS FOR UNMANNED AIRCRAFT SYSTEMS

Session Chair : Prof. Ying-Chih Lai

Time	Paper ID	Title & Author
10:30-10:45	**#1143**	**Development of an Automatic Launcher for Fixed-Wing UAVs and a Multi-UAV Intelligent Cooperative Path Planning Method** Meng-Tse Lee, Cheng-Han Zhuang, Yi-Le Chen
10:45-11:00	**#1170**	**A 3-D Robust Hybrid Time-of-Arrival Simulation Study for Fixed Terrestrial Base Stations with Dual-Band RF Propagation** Ting-Yu Su, Ting-Yi Li, Jyh-Ching Juang
11:00-11:15	**#1198**	**PAUSE-Assisted Ambiguity Validation for Dual-Frequency iGPS-RTK in Outdoor Environments** Ting-Yi Li, Ting-Yu Su, Jyh-Ching Juang
11:15-11:30	**#1253**	**Does Rotor-Drag Modeling Change the Control Strategy Learned by Reinforcement Learning?** Huan-Zheng Su, Woei Leong Chan

CACS-SS-26 **(Room V008)** Friday, Oct. 23, 2026, 10:30 ~ 12:00

TOPIC: ADVANCED GUIDANCE, NAVIGATION AND CONTROL SYSTEMS FOR UNMANNED AIRCRAFT SYSTEMS

Session Chair : Prof. Ying-Chih Lai

Time	Paper ID	Title & Author
10:30-10:45	**#1143**	**Development of an Automatic Launcher for Fixed-Wing UAVs and a Multi-UAV Intelligent Cooperative Path Planning Method** Meng-Tse Lee, Cheng-Han Zhuang, Yi-Le Chen
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11:00-11:15	**#1198**	**PAUSE-Assisted Ambiguity Validation for Dual-Frequency iGPS-RTK in Outdoor Environments** Ting-Yi Li, Ting-Yu Su, Jyh-Ching Juang
11:15-11:30	**#1253**	**Does Rotor-Drag Modeling Change the Control Strategy Learned by Reinforcement Learning?** Huan-Zheng Su, Woei Leong Chan
11:30-11:45	**#1268**	**Decentralized Dynamic Task Assignment of Swarm UAV Systems for Search and Rescue Missions** Chang-Lin Hsieh, Ying-Chih Lai

CACS-SS-29 **(Room V009)** Friday, Oct. 23, 2026, 10:30 ~ 12:00

TOPIC: OBSERVER-BASED CONTROL DESIGN FOR MECHATRONIC SYSTEMS

Session Chair : Prof. Ming-Yang Cheng

Time	Paper ID	Title & Author
10:30-10:45	**#1114**	**A Practical Voltage-Model-Based Sensorless FOC Implementation for High-Speed AI Server Fan Drives** Jun-Ping Chang, Ming-Yang Cheng
10:45-11:00	**#1130**	**Neural Network-Based Robotic Dynamics Modeling and PSO-Based Identification Trajectory Design for a 2-DOF Robot Manipulator** Chi-Shan Hung, Yi-Ming Tien, Yi-Cheng Hsiao, Ping-Huan Cheng, Ming-Yang Cheng
11:00-11:15	**#1145**	**A Variable-Gain Nonlinear Flux Observer for PMSM Server Fan Drives: Low-Speed Estimation and High-Speed Sensitivity to Stator Resistance Mismatch** Jun-Ping Chang, Kuan-Yu Lee, Ming-Yang Cheng
11:15-11:30	**#1161**	**H2 and H∞ Observer Designs for Disturbance Estimation in Motor Speed Control** Chi-Yang Chang, Syu-Ruei Jhang, Mi-Ching Tsai

Session Track 5

Oct. 23, 2026, 13:00 ~ 14:30

CACS-R-02 **(Room V001)** Friday, Oct. 23, 2026, 13:00 ~ 14:30

TOPIC: ADVANCED CONTROL TECHNOLOGIES AND APPLICATIONS

Session Chair : Prof. Chan-Yun Yang

Time	Paper ID	Title & Author
13:00-13:15	**#1071**	**Small-Sample Region-Aware Void Segmentation via a Learned Inspection Region for Aerospace Aluminide Coatings** Peng-Wen Lai, Hao-Chun Pan, Jau-Woei Perng, Bing-Fei Wu
13:15-13:30	**#1173**	**Kinematic Modeling and Lateral Drift Compensation for a T-Shaped Heterogeneous Omnidirectional Mobile Robot** Yin-Jui Liao, Chia-Ti Wu
13:30-13:45	**#1196**	**AIoT-Oriented Impact-Sound Classification with Cross-Model Decision-Level Fusion for Smart Plastic Recycling** Tsun-Hsin Chang, Ting-Cheng Yeh, Yu-Hao Lin
13:45-14:00	**#1197**	**A Non-Contact Blood Glucose Estimation Approach from Facial Images via Remote Photoplethysmography** Jhih-Sian Lin, Wei-Peng Tang, Bing-Fei Wu
14:00-14:15	**#1261**	**Bimodal Emotion Recognition Based on Lightweight Acoustic Features and Pre-trained Language Models** Chang Pei-Fen, Shi Yan-Ren, Shing-Tai Pan

CACS-SS-04 **(Room V002)** Friday, Oct. 23, 2026, 13:00 ~ 14:30

TOPIC: DATA-DRIVEN COMPUTATIONAL INTELLIGENCE TECHNIQUES AND APPLICATIONS

Session Chair : Prof. I-Fang Chung

Time	Paper ID	Title & Author
13:00-13:15	**#1023**	**Comparison of Federated Learning Strategies for Machine Learning** Shih-Cheng Horng, Shieh-Shing Lin, Yin-You Lai
13:15-13:30	**#1097**	**Indoor Fall Detection Using Deep Learning-based Posture Classification and Temporal Posture Sequence Decision** Jung-Yu Chang, Tao-Shuo Wang, Chia-Feng Juang
13:30-13:45	**#1116**	**End-to-End Portfolio Optimization with Hybrid Transformers and Differentiable Convex Optimization** Ti-Chung Lee, Chun-Yu Chou
13:45-14:00	**#1166**	**Vision-based Indoor Mobile Robot Navigation Using Supervised Deep Learning Network with a Monocular Camera** Chia-Feng Juang, Chao Bo-Wei
14:00-14:15	**#1251**	**Cancer-Type-Aware Adversarial Domain Adaptation for Cell-Line- to-Patient Drug Response Prediction** Chi-Tang Wang, Yu-Kai Chiu, I-Fang Chung
14:15-14:30	**#1295**	**VisionBag: A Gaze-Guided Backpack-Mounted Assistive Caption Display Using Fuzzy-PID Visual Servoing** Hong-Ru Chen, Rong-Sheng Chen, Jin-You Hong, Cheng-En Li, Cheng Hsien Lin, Tzu-Hong Yang, Yi-Wei Chen, You-en Hsu, Chiun-Li Chin

CACS-SS-09-01	**(Room V003)** Friday, Oct. 23, 2026, 13:00 ~ 14:30	
TOPIC: AI BASED ROBOTICS		
Session Chair : Prof. Wen-June Wang		
Time	**Paper ID**	**Title & Author**
13:00-13:15	**#1073**	**An Interactive Robot System Integrating with Mixed Reality, Digital Twin and Multilingual Dialogue Models** Hsuan Wei Wu, Han-Pang Huang
13:15-13:30	**#1088**	**Multi-Label Skeleton-Based Action Recognition for Elderly Care Service-Need Inference** Ya-Ting Lee, Sheng-Kai Lee, Hsien-Wei Chiu, Chian-Song Chiu
13:30-13:45	**#1184**	**Design and Implementation of a Generative AI Agent-Based Interactive English Teaching Robot** Guan-Zhen Chen, Bo-Wei Shih, Rong-Jyue Wang, Zhen-Xuan Hong
13:45-14:00	**#1271**	**SLAM2Sim: From Sparse SLAM Point Clouds to Simulation-Ready 3D Scenes for Robotics** Jui-Yun Tsai, Siang-Yeh Ho, Chih-Yang Lin
14:00-14:15	**#1287**	**Integrating Learned Corrections into Sequential Neural Inverse Kinematics: Residual Learning, Distillation, or Co-Training?** Chun-Chih Chang, Yi-Ling Tsai, Yi-Ming Huang, Si-Ying Yu, Po-Yen Wu, Po-Chiang Lin

CACS-SS-23-01 **(Room V004)** Friday, Oct. 23, 2026, 13:00 ~ 14:30

TOPIC: DEVELOPMENT AND INTEGRATION OF INTELLIGENT MEDICAL DIAGNOSIS AND REHABILITATION CARE SYSTEMS

Session Chair : Prof. Chih-Min Lin

Time	Paper ID	Title & Author
13:00-13:15	**#1075**	**Signal Analysis and Feature-Based Machine Learning for Arteriovenous Graft Stenosis Detection** Yun-Chia Chen, Yi-Chieh Yu, Wei-Ling Chen, Yu-Te Wu, Chi-Wen Jao
13:15-13:30	**#1125**	**Toward Explainable Gait Assessment in Smart Healthcare: Insights from Post-Stroke Gait Kinematics** Meng-Lin Tsai, Po-Chuan Chiou, Jing-Chen Hong
13:30-13:45	**#1194**	**Artificial Intelligence Techniques for Optical Coherence Tomography Images in Vitreomacular Interface Abnormalities** Cheng-Hung Dixson Lin, Pin-Han Wen, Chih-Ling Tseng, Jia-Kang Wang, Tzulun Huang
13:45-14:00	**#1195**	**Few-Trial Personalized Left–Right Lower-Limb EEG Decoding for Stroke Patients Using a Pretrained Multi-Scale Feature Extractor** Shiyong Luo, Yu-Wei Lin, Cong-Ying He, Yuan-Yang Cheng, Chen Chia-Hsin, Po-Hsun Cheng, Li-Wei Ko
14:00-14:15	**#1229**	**Fast 4D Human Mesh Reconstruction for VR Head-Mounted Displays** Yu-Chen Cheng, Yu-Hui Huang
14:15-14:30	**#1231**	**Frequency-Dependent Spectral Characteristics and Temporal Stability of Steady-State Visual Evoked Potentials during Sustained Stimulation** Kun-Wei Liu, Chia-Yi Chou, Chih-Tsung Chang, Yi-Chieh Hsu

CACS-SS-18-02 **(Room V005)** Friday, Oct. 23, 2026, 13:00 ~ 14:30

TOPIC: AI-EMPOWERED SMART DRIVING TECHNOLOGY

Session Chair : Prof. Shih-Hsien Hsu

Time	Paper ID	Title & Author
13:00-13:15	**#1038**	**Development of an Obstacle Avoidance System for Autonomous Vehicle using Reinforcement Learning and PID Control** Pisanu Palasin, Tanagorn Jennawasin
13:15-13:30	**#1109**	**VLA-based road-following technology applied to JetBot** Jen-Hsing Li
13:30-13:45	**#1187**	**A Two-Stage Real-Time Bridge Collision Avoidance Framework for Overheight Vehicle Navigation** Cheng-Yu Hsieh-Lu, Jen-Chun Lee, Chia-Ti Wu, Yao Wan-Ru
13:45-14:00	**#1189**	**Common-Mode Braking Strategy for Anti-lock Braking System and Electronic Stability Control under Combined Slip Conditions** Jia-Le Wei, Tsung Hua Hsu, Yan Xiang Zhang
14:00-14:15	**#1259**	**Trajectory Planning for Safety-Triggered Automatic Lane-Change Maneuvers: Hardware-in-the-Loop Validation** Hsin Han Chiang, Chi-Hai Huang, Mei-Lin Huang

CACS-SS-19-02 **(Room V006)** Friday, Oct. 23, 2026, 13:00 ~ 14:30

TOPIC: INTELLIGENT CONTROL IN ROBOTICS AND ENERGY APPLICATIONS

Session Chair : Prof. Kai-Yi Wong

Time	Paper ID	Title & Author
13:00-13:15	**#1059**	**A GFM/GFL-Based Microgrid Platform for Stability and Resilience Assessment** Po-Yuan Wu, Yi-Quan Liao, Ting-Yue Chen, Sei Zhen Khong, Kuo-Sheng Hsieh, Wen-Zhuang Jiang, Hung-Wei Cheng, Ching-Jung Liao
13:15-13:30	**#1101**	**Flight Control Performance Comparison of 3D-Printed vs. Commercial UAV Propellers** Hui-Hsuan Lo, Yu-Hsien Chu, Wei-Chun Lin, Kai-Yi Wong
13:30-13:45	**#1215**	**Experience Fusion Network for Distributed Multi-Agent Deep Reinforcement Learning** Jui-Yao Wu, Rong-Pei Chuang, Wei-Cheng Jiang
13:45-14:00	**#1269**	**Charging Task Allocation with Bayesian Networks in Warehouse** Ching-Wei Chuang, Bing-Wei Hua
14:00-14:15	**#1286**	**Integral Sliding Mode Control for Trajectory Tracking of Two-Wheeled Self-Balancing Robots** Hsuan-Yang Shen, Jenq-Lang Wu

CACS-SS-25 **(Room V007)** Friday, Oct. 23, 2026, 13:00 ~ 14:30

TOPIC: ADAPTIVE AND LEARNING CONTROL OF INTELLIGENT ROBOTIC SYSTEMS

Session Chair : Prof. Ming-Li Chiang

Time	Paper ID	Title & Author
13:00-13:15	**#1037**	**Humanoid Robot Locomotion Control via Privileged State Estimation and Self-Supervised Learning** Chia-Cheng Hsin, Hou-Shin Wu, Tzuu-Hseng S. Li
13:15-13:30	**#1211**	**Integration of Safe Adaptive Variance Exploration and MPC-Based Deterministic Policy Gradient for UAV Trajectory Optimization** Yu-Chun Ting, Che-Yu Chou, Ming-Li Chiang
13:30-13:45	**#1236**	**Pose-Aware Confidence-Guided Environment Representation and Safety Refinement for Limited-FOV Quadruped Navigation** Yiru Ye, Feng-Li Lian
13:45-14:00	**#1250**	**Design of Curvature-Assisted Safety Planning for Large Vehicles Using Model Predictive Control** Bo-Fan Chen, Hsiang-Yu Wen, Ming-Li Chiang
14:00-14:15	**#1289**	**Video Frame Interpolation Lightweight Network based on Multi-Frame Optical Flow Estimation History** You-Xun Guo, Cheng-Ming Huang

CACS-SS-27 **(Room V008)** Friday, Oct. 23, 2026, 13:00 ~ 14:30

TOPIC: REAL-TIME INTELLIGENT SYSTEMS AND CONTROL FOR ADVANCED MECHATRONICS

Session Chair : Prof. Nien-Tsu Hu

Time	Paper ID	Title & Author
13:00-13:15	**#1054**	**Saturation-Aware Actor–Critic Optimal Tracking Control of a Single-Link Robotic Manipulator** Nien-Tsu Hu, Hsiang-Tung Kao, Chin-Sheng Chen
13:15-13:30	**#1102**	**Real-Time Workplace Safety Detection Using Vision-Language Models** Ly Tri, Chin-Sheng Chen
13:30-13:45	**#1106**	**Finite-Time Stabilization of a Fractional-Order Lü Chaotic System** Chih-Cheng Huang, Cheng-Yi Chen, Fu-Ming Chen, Zong-Yu Hong, Guan-Zong Wang, You-Ting Xie, Ming-Hung Lin
13:45-14:00	**#1126**	**Order-Dependent Dynamic Transitions in a Fractional-Order Chen Chaotic System** Ming-Hung Lin, Cheng-Yi Chen, Sheng-Han Wu, Guan-Zong Wang, Shuo-Tsung Chen, You-Ting Xie, Chih-Cheng Huang
14:00-14:15	**#1206**	**Design of a Low-Power Transimpedance Amplifier Integrated Circuit for Optical Receiver Applications** Chih-Lung Tseng, Tsung-You Wu, Chan-I Lin

CACS-SS-31-01 **(Room V009)** Friday, Oct. 23, 2026, 13:00 ~ 14:30

TOPIC: ADVANCED CONTROL AND ANALYSIS FOR INTELLIGENT MECHATRONIC SYSTEMS

Session Chair : Prof. Shyh-Leh Chen

Time	Paper ID	Title & Author
13:00-13:15	**#1078**	**Hybrid Friction Modeling and Simultaneous Robot Parameter Identification via Separable Nonlinear Regression** Chih-Hsien Li, Woraphrut Kornmaneesang, Jiravit Pratvittaya
13:15-13:30	**#1112**	**Data-Driven Iterative Learning Contouring Control** Thanh-Quan Ta, Shyh-Leh Chen
13:30-13:45	**#1163**	**Optimal H2 Control Design for Model-Matching with Enhanced Disturbance Rejection** Chi-Yang Chang, Cheng-Chung Tien, Mi-Ching Tsai
13:45-14:00	**#1178**	**A Lightweight YOLO11 Variant for Citrus Canker Lesion Detection under Natural Field Conditions** Tzyy-Chyang Lu, Kai-Chun Chang
14:00-14:15	**#1179**	**Dynamics-aware Motion Planning for multi-axis motion systems** Ejaz Ahmad, Shyh-Leh Chen
14:15-14:30	**#1180**	**Towards the Implementation of a Smooth Motion Control System for Robotic Manipulators Using the EtherCAT Protocol** Yu-Sheng Lu, Wei-Lun Hong, Chih-Wen Lo

Session Track 6

Oct. 23, 2026, 14:50 ~ 16:05

CACS-SS-32 **(Room V001)** Friday, Oct. 23, 2026, 14:50 ~ 16:05

TOPIC: BRAINSYNERGY: AN OPEN-SOURCE AI PLATFORM FOR MULTIMODAL NEUROIMAGING AND APPLICATIONS ACROSS THE LIFESPAN

Session Chair : Prof. Jeen-Shing Wang

Time	Paper ID	Title & Author
14:50-15:05	**#1135**	**Deep Phenotyping of CADASIL for Precision Medicine in Cerebral Small Vessel Disease** Yuan Ting Sun
15:05-15:20	**#1188**	**AIoT Classroom Lighting Modulates Anterior Cingulate Excitation-Inhibition Balance and Evening Melatonin in Female High School Students** Tsung-Hua Lu
15:20-15:35	**#1234**	**Validation of Cerebral Blood Flow Quantification Using Radiologist MRI Injury Scoring in Neonatal Hypoxic-Ischemic Encephalopathy** Pei Chun Lai, Yi-Fang Tu
15:35-15:50	**#1248**	**BrainSynergy: An Open-Core Platform for Development-Aware MRI Analysis and Traceable MRS Workflow Validation** Tzu-Yu Lei, Jing-Yuan Huang, Hsuan-Kai Huang, Hsin-Hsien Chen, Jeen-Shing Wang
15:50-16:05	**#1283**	**Classroom-Based AIoT Human-Centric Lighting and Mindfulness Training for Attention Regulation in Female High School Students** Ya-Ting Carolyn Yang, Yun-Chen Lin, Jamie Zeitzer, Tsung-Hua Lu, Yen Kuang Yang, Hsin-Hsien Chen, Jeen-Shing Wang

CACS-SS-12-02 **(Room V002)** Friday, Oct. 23, 2026, 14:50 ~ 16:05

TOPIC: AI-DRIVEN PERCEPTION, PREDICTION, AND DECISION-MAKING FOR AUTONOMOUS SYSTEMS

Session Chair : Prof. Wei-Lun Lin

Time	Paper ID	Title & Author
14:50-15:05	**#1151**	**End-to-End Automotive Camera and Heterogeneous AI Integration on TI TDA4VM for Real-Time Vulnerable Road User Warning** Wang Pao-Kai, Hong-Yu Chen, Yu-Chen Lin
15:05-15:20	**#1159**	**Heterogeneous Sensor Integration for Autonomous Navigation and Pallet Docking on an Ackermann-Steered Forklift Prototype** Shi-Yu Hong, Ching-Ho Wu, Huang Ze-ming, Yu-Chen Lin
15:20-15:35	**#1192**	**Field-Based Multi-Sensor Condition Monitoring of Aviation Electromechanical Actuators Using Cross-Platform Validation** Sheng Tao Chen
15:35-15:50	**#1266**	**Fixed-Budget Margin-Based Semantic Consistency Filtering for Generative Generalized Zero-Shot Learning** Yu-Chuan Lin, Wei Lun Lin
15:50-16:05	**#1294**	**A Multi-Feature Correlation Filter Tracker Combined with Optical-Flow-Based Distance Estimation for Unmanned-Vehicle Visual Following** Kaileng Ciou, Yi-Ting Yu, Ao-Hsi Kuo, Kuang-Yow Lian

CACS-SS-09-02	**(Room V003)** Friday, Oct. 23, 2026, 14:50 ~ 16:05	
TOPIC: AI BASED ROBOTICS		
Session Chair : Prof. Wen-June Wang		
Time	**Paper ID**	**Title & Author**
14:50-15:05	**#1085**	**Reinforcement Learning for Robotic Flower Insertion Tasks Based on Multi-Stage Reward Shaping** Hao-Yang Chen, Yu-Ping Liao, Chian-Song Chiu
15:05-15:20	**#1129**	**A Non-Contact Blood Glucose Estimation Approach from Facial Images via Remote Photoplethysmography** Jhih-Sian Lin, Wei-Peng Tang, Bing-Fei Wu
15:20-15:35	**#1132**	**Improving Limited-Data Image Classification Accuracy with a Two-Stage Augmentation Strategy by GAN-Based and Linear Mixups** Chih-Hsiu Lu, Yi-Cheng Wang, Shao-Ting Wen, Jui-Chieh Yu, I-Chieh Chiu, Jim-Wei Wu
15:35-15:50	**#1181**	**Human–Robot Object Handover Planning Using Multimodal Semantic Reasoning and Affordance Analysis** Hsiang-Chieh Chen
15:50-16:05	**#1284**	**Reinforcement Learning for Control of TIAGo Robot Manipulator Arm: A Comparative Study of Actor-Critic Methods and Motion Planning** Jim Torresen, Markus Toverud Ruud

CACS-SS-23-02 **(Room V004)** Friday, Oct. 23, 2026, 14:50 ~ 16:05

TOPIC: DEVELOPMENT AND INTEGRATION OF INTELLIGENT MEDICAL DIAGNOSIS AND REHABILITATION CARE SYSTEMS

Session Chair : Prof. Jing-Chen Hong

Time	Paper ID	Title & Author
14:50-15:05	**#1074**	**Continuous Noninvasive Physiological Monitoring for Comparative Assessment of Kangaroo Care in Preterm Infants** Huan-Hsu Chen, Teh-Ming Wang, Chuang-Chien Chiu, HsinHung Kuo
15:05-15:20	**#1115**	**A Lightweight Deep Learning Model for Breast Lesion Classification** Vo Trong Quang Huy, Lo-Yi Lin, Chih-Min Lin
15:20-15:35	**#1128**	**Microphone-Robust Pathological Voice Classification** Syu-Siang Wang, Chao-Hsiang Hung
15:35-15:50	**#1171**	**Development and Design of a Wheelchair-Pushing Robot System** Ying Jen Chen, Tsung-Yi Thish

CACS-SS-03 **(Room V005)** Friday, Oct. 23, 2026, 14:50 ~ 16:05

TOPIC: NEXT-GENERATION ROBOTIC PERCEPTION: INTEGRATING DEEP LEARNING AND MACHINE VISION

Session Chair : Prof. Yi-Zeng Hsieh

Time	Paper ID	Title & Author
14:50-15:05	**#1028**	**Geometric Consistency Verification for YOLO Detections under Occlusion Using RGB-D Depth Variance** Tung-Min Chang, Hsing-Cheng Yu
15:05-15:20	**#1174**	**Diagnostic-Driven Video Mapping and Dual-Layer Localization** Hung Ching Lun, Yi-Zeng Hsieh
15:20-15:35	**#1216**	**CNN-Based Image Localization for Indoor GPS-Denied Smart Parking Path Navigation** Ching-huei Huang, Yi-Ching Huang, Ting-En Chen, Kao Huan Hsun, Tz-Heng Hung
15:35-15:50	**#1224**	**UAV Real Time Path Localization Using SuperPoint–LightGlue Visual Positioning with IMU Compensation** Ching-huei Huang, Yu-Hao Chen, Pei-Yu Liao, Chong-Qin Liao, Yi-Tong Chen
15:50-16:05	**#1240**	**Recognition-Oriented Super-Resolution for Resource-Constrained Underwater Visual Perception** Dong-Lin Li, Shin-He Wu

CACS-SS-07 **(Room V006)** Friday, Oct. 23, 2026, 14:50 ~ 16:05

TOPIC: AUTOMATIC CONTROL FOR HUMAN-ROBOT COLLABORATION

Session Chair : Prof. Po-Ting Lin

Time	Paper ID	Title & Author
14:50-15:05	**#1012**	**Vision-Language-Action Control for Autonomous Mobile Robots Using Retrieval-Augmented Language Reasoning** Lin-Shang Pan, Chii Hong Liu, Po Ting Lin
15:05-15:20	**#1060**	**Reflector-aided Cartographer Simultaneous Localization and Mapping (RCSLAM) for a High-payload Autonomous Mobile Robot (HAMR)** Chen-Fei Yang, Yi-Chuan Tseng, Po Ting Lin
15:20-15:35	**#1062**	**Collision-Avoiding Path Planning of a Dual-Arm Robotic System Based on Augmented Lagrangian Minimization** Nai-Hsiang Hung, Zi-Xuan Cheng, Po Ting Lin, Yu-Hsun Chen
15:35-15:50	**#1121**	**Feature Preservation and Adaptive Fusion for Small Object Detection in UAV Imagery** Ting-Ling Ku, Hui Che Hsu, Chien-Ming Huang, Auzan Izzaturrahman, Ching-Chan Liao, Shang-Lin Li, Yung-Yao Chen
15:50-16:05	**#1213**	**Programmable PWM-Based Non-Stationary Gear Excitation System** Sih-Hao Huang, Chia-Hung Lai, Shu-Hsien Huang

CACS-SS-28 **(Room V007)** Friday, Oct. 23, 2026, 14:50 ~ 16:05

TOPIC: DESIGNS AND APPLICATIONS OF GAI BASED SERVICE ROBOTS

Session Chair : Prof. Tzuu-Hseng S. Li

Time	Paper ID	Title & Author
14:50-15:05	**#1034**	**Design, Implementation, and Control of a Bimanual Service Robot for Embodied AI Applications** Syuan-Rong Chen, Hsiang-Cheng Yu, Ci-Dong Cheng, Ting-Yu Jiang, Wen-Huang Chen, Chong-En Wang, Yi-Chieh Hsueh, Zih-Ming Su, Tzuu-Hseng S. Li
15:05-15:20	**#1035**	**Integrating Self-Correcting Cognitive Task Planning with Hybrid Visual Localization for Bimanual Robots** Syuan-Rong Chen, Chong-En Wang, Tzuu-Hseng S. Li
15:20-15:35	**#1043**	**Vision-Language-Action Model for Manipulations** Yan-Zong Wu, Chen-Wen Chang, Po-Hsun Feng, Her-Terng Yau, Tzuu-Hseng S. Li, Ping-Huan Kuo
15:35-15:50	**#1080**	**Vision-Language Model-Based Human-Following Strategies for Companion Robots** Cong-Thanh Vu, Yen-Chen Liu
15:50-16:05	**#1255**	**Fine-Tuning Whisper for Robust Speech Recognition in Older Adults with Cognitive Impairment** Ya-Ning Chang, Tzuu-Hseng S. Li, Xiang-Yi Zhang

CACS-SS-30 **(Room V008)** Friday, Oct. 23, 2026, 14:50 ~ 16:05

TOPIC: PERCEPTION AND DECISION-MAKING OF INTELLIGENT SYSTEMS

Session Chair : Prof. Chen-Chien Hsu

Time	Paper ID	Title & Author
14:50-15:05	**#1069**	**Knowledge-Grounded Unified Reasoning for Anomaly Handling and Spoken-Command Translation in Hospital Patrol Robots** Chao-Chun Liang, Chun-Hung Chen, Wen-Lian Hsu
15:05-15:20	**#1131**	**A Lightweight Edge Framework for Real-Time Multi-Person Action Recognition in Older-Adult Care** Hoangcong Le, Cheng-Kai Lu, Wei-Yen Wang, Chen Chien Hsu
15:20-15:35	**#1165**	**Psychological Counseling Support System: Stepwise Reasoning Based on LLMs** Min-Sheng Weng, Min-Jie Hsu, Kai-Yang Hsueh, Ying-Chun Li, Ting-Yu Lin
15:35-15:50	**#1168**	**A System for Generating Control Signals for Vision-Guided Robots Based on Video Features** Zhi-Huan Chen, Min-Jie Hsu, Yu-Zhen Wu
15:50-16:05	**#1223**	**An Edge-Based Voice RAG Kiosk for Hospital Health Education** Lee Hsien-Wen, Peng Cheng-Wei, Yi-Hsing Chien, Wei-Yen Wang

CACS-SS-31-02 **(Room V009)** Friday, Oct. 23, 2026, 14:50 ~ 16:05

TOPIC: ADVANCED CONTROL AND ANALYSIS FOR INTELLIGENT MECHATRONIC SYSTEMS

Session Chair : Prof. Woraphrut Kornmaneesang

Time	Paper ID	Title & Author
14:50-15:05	**#1051**	**Locally Semiconcave Control Lyapunov Function for Time-varying Systems and Application to Terminated Trajectory Tracking** Kento Okubo, Hisakazu Nakamura
15:05-15:20	**#1103**	**Development of Prescribed-Time Motion Control for Four-Wheeled Mecanum Omnidirectional Mobile Robots** Hsu-Chih Huang
15:20-15:35	**#1208**	**A Novel Flywheel Energy Storage System with Low Energy Consumption** Yun-Sih Lin, Shyh-Leh Chen
15:35-15:50	**#1237**	**Smooth Raster-Like Trajectory Design and Adaptive Neural-Network Control for Increasing AFM Scanning Speed** Huang-Chih Chen, Li-Chen Fu
15:50-16:05	**#1272**	**Sensitivity of SST Gap-Filling Performance to Cloud-Mask Geometry and Persistence** Teerathep Yaisungnoen, Jiravit Pratvittaya

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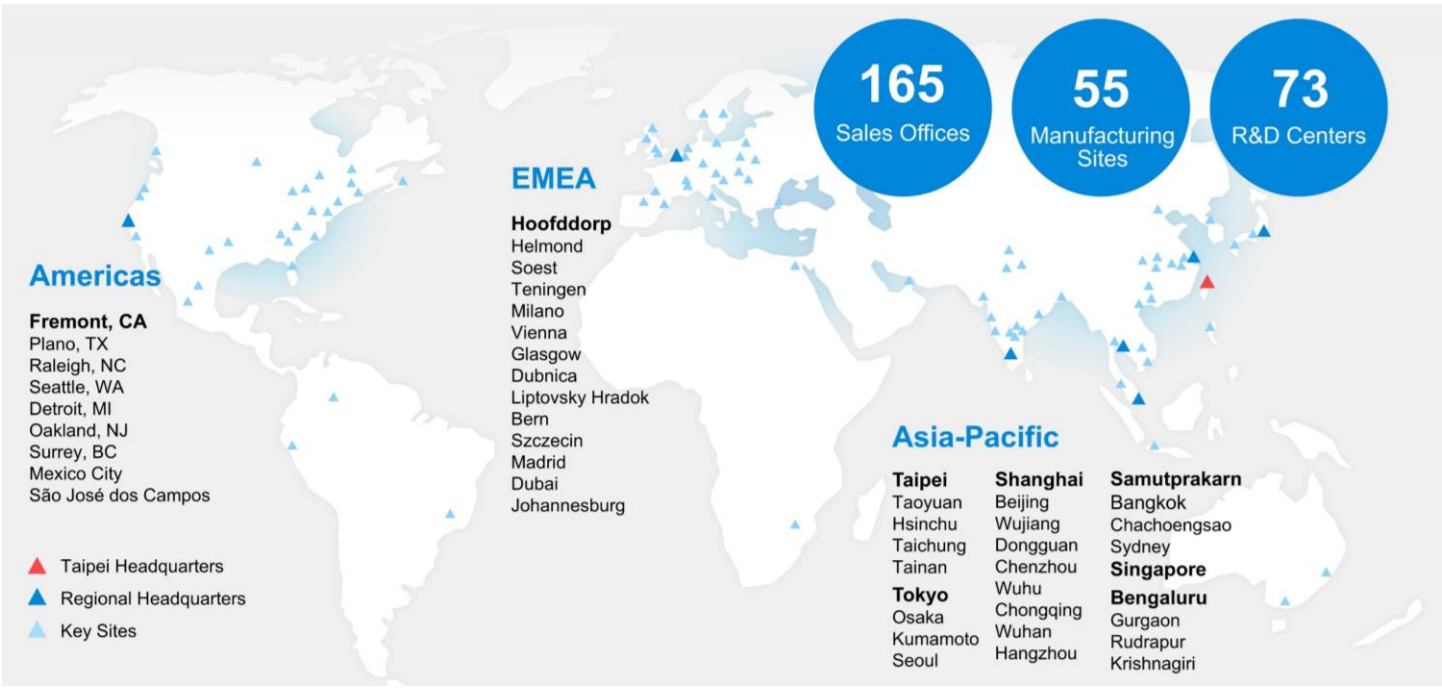

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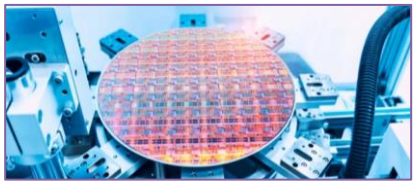

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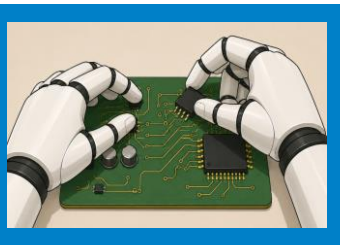

Technology Focus

Inputs	Sensing & Reasoning			Acts	
Data Collection	Tactile Sensing	Model Training	Behavior Planning	End Effector / Hand	Configuration Modularity

Talent Requirement

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- Robotics Software Architecture & Systems Engineer
- Robotics R&D Engineer
- Robot Learning Engineer
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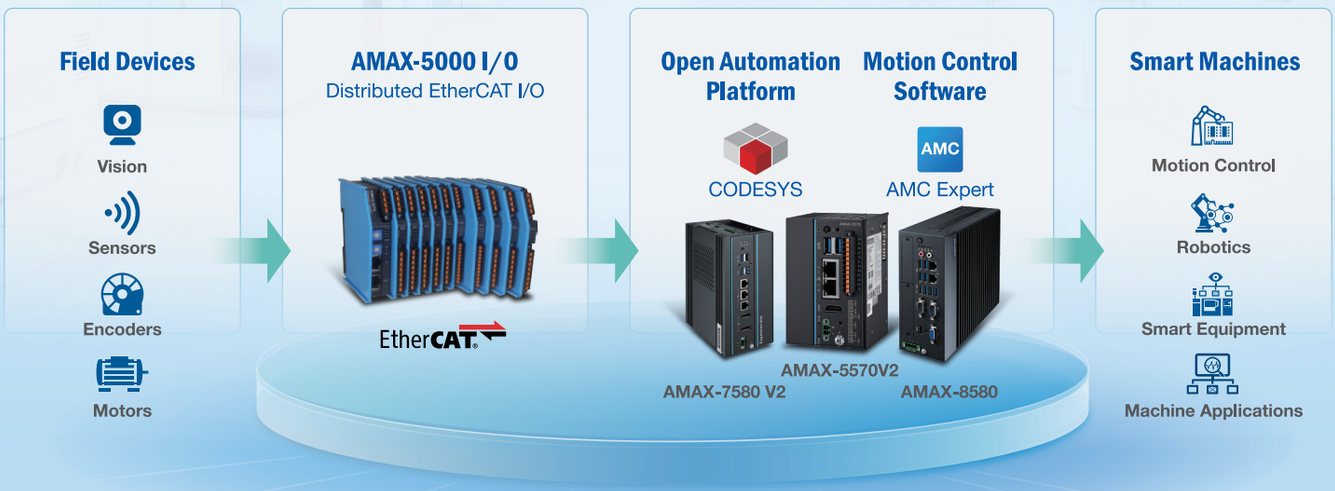

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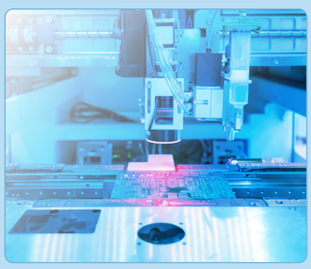
APPLICATIONS

Semiconductor Equipment



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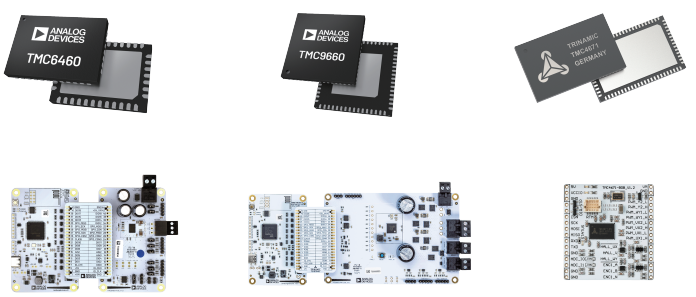

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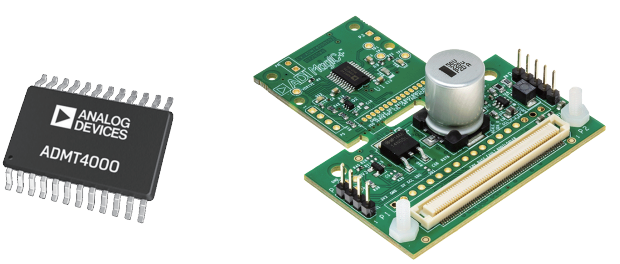

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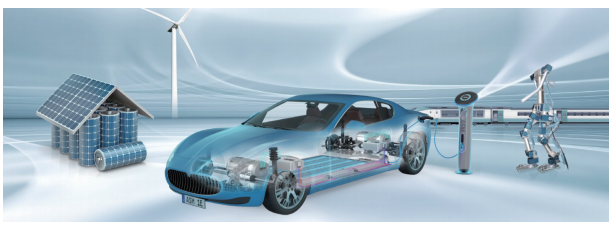

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Origin 提供科學家與工程師所需的強大數據分析能力與印刷品等級的繪圖功能。Origin 從使用者觀點出發的介面設計，根據科學家以及工程師的需求，將數據資料轉換成各種圖型的最佳解決方案，包含圖型繪製、數學統計、曲線擬合、峰值分析和訊號處理，並可以透過程式編寫開發，將您的軟體因應多元的資料進行客製化！

WOLFRAM MATHEMATICA 15

高階數學及符號運算軟體

Mathematica 在科學計算領域確立了最先進的技術，為全球技術創新人員、教育工作者、學生和其他人士提供了最主要的計算環境，並成為許多機構數學運算的標準工具。Mathematica 以卓越的技術和簡便的使用方法享譽全球，它提供了單個集成並且持續擴展的系統，不僅用於計算，也能用於建模、模擬、視覺化、軟體開發、建立文件或是輸出成果，涵蓋了最廣最深的科學計算功能。

S F I 昊青股份有限公司 SCIENTIFIC FORMOSA, INC.
免費服務專線 0800-099-900
www.sciformosa.com.tw

PAPAGO!® FACE8 台灣臉霸

昆蟲/動物影像辨識

手機軟體或攝影機自動偵測拍照，上傳雲端後台儲存影像並運用AI進行物種辨識，支援編輯與分析，減輕人工作業負擔。

AI聊天機器人

整合跨系統資料庫，利用AI與自然語言處理技術，支援內部資訊檢索與統計分析，提供景點導覽及園區資訊諮詢。

動態車牌辨識

結合高快門攝影機與AI影像辨識技術，車輛高速行駛或光線不足時，即時擷取並辨識車牌號碼。

門禁/差勤管理

可依需求彈性搭配口罩身分辨識、體溫量測與活體檢測功能。

工安防護系統應用

人臉辨識工地管理

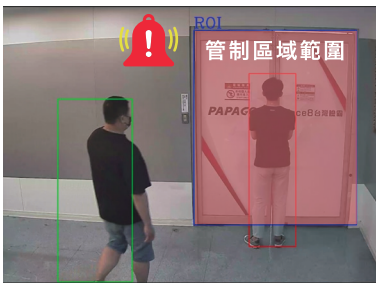
管理所有施工團隊現場的進出狀態，系統後台可同步監控人流，強化工地安全管理，降低意外發生風險。

微氣象站

即時監測環境數據，涵蓋危害指數、危害等級、溫/濕度、風速、風向、PM2.5、PM10、噪音與均能音量，協助工地或戶外作業掌握氣候變化與環境狀況。

電子圍籬

提供危險範圍警示功能，偵測範圍可設定不規則形狀，支援電子圍籬與遠端ROI設定，並即時將警報通知推播給指定人員。



人員裝備辨識

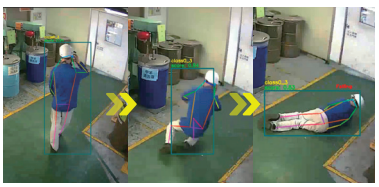
透過AI影像防護裝備識別功能，檢查人員是否依照場域指示穿戴適當的防護裝備，例如安全帽、反光防護背心等。如未依規定穿著裝備，系統將發出警示通知。

人機協作辨識管理

即時偵測作業人員是否位於安全範圍內；若偵測到人員進入危險區域，系統將立即停止或限制機具運作，以降低作業風險。

人員行為辨識

辨識人員於作業期間發生的跌倒、突發異常行為等狀況，並主動通知管理人員，協助現場迅速應變與處理。



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官方網站

實績案例

Orange Electronic

TPMS SYSTEM

橙的電子 (Orange Electronic Co., Ltd.) 股票代號：4554
專注於無線胎壓監測系統 (Tire Pressure Monitoring System, TPMS) 技術研發、生產製造、全球行銷、品牌經營的跨國企業，總公司位於台灣中部科學園區，我們國際化的經營戰略深耕台灣、放眼世界，在美國、德國、荷蘭、義大利、蘇州...等地皆設有公司，能夠快速服務世界各地客戶。

Orange Electronic 成立以來深耕自有品牌、堅持產品品質、自主研發設計、發展國際化技術，多次榮獲各項大獎殊榮：臺灣精品獎、國家磐石獎、小巨人獎、中華汽車 Top Quality 評鑑優良供應商...等肯定，已是國內外各大原廠車廠標配/選配/Aftermarket/輪胎廠的主要合作夥伴，在TPMS領域中取得全球領先優勢。自有國際品牌及國內外車廠一階供應商同時發展，提供國際一流時尚設計辦公環境，世界級台灣汽車電子精品。

JOIN US

Orange Electronic

● 研發工程師
【 資料庫維護工程師 】

NEW

職務說明

(1.) 熟Python，用於開發數據處理、自動化工具以及腳本化解決方案，增強專案靈活性與效率。
(2.) 具有資料庫概念者尤佳。(3.) 其他主管交代事項。

● 研發工程師
● 助理工程師
【 韌體工程師 】

NEW

職務說明

(1.) 有使用C#撰寫測試程式的經驗。 (2.) 有CAN的know how尤佳。
(3.) 有MCU使用經驗尤佳。
(4.) 具備良好的編碼習慣，包括代碼 可讀性、模組化和註釋。
(5.) 能夠獨立進行測試，具備故障排除和問題解決的能力，以確保系統的穩定性和可靠性。
(6.) C#開發經驗，有較高的技術能力。

● 研發工程師
● 助理工程師
【 Android開發設計 】
【 iOS APP開發設計 】

NEW

職務說明

(1.) 熟悉 Kotlin 語言，具備 Android 應用開發經驗。
(2.) 熟悉 GitHub 版控，具備良好的溝通能力，能夠進行團隊協作。
(3.) 熟悉原生系統與 JS 交互開發模式尤佳 (WebView)。
(4.) 具備 Web 開發經驗尤佳 (HTML、JavaScript、CSS)。
(5.) 具備藍芽、UART 通訊開發經驗尤佳。
(6.) 能夠獨立進行測試，具備故障排除和問題解決的能力，以確保系統的穩定性和可靠性經驗尤佳。

● 資訊工程師
● 助理工程師
【 前端工程師 】

NEW

職務說明

(1.) 前端技術建構網站頁面。
(2.) 與後端工程師協調進行API串接，確保網站畫面與資料庫的順暢連接。
(3.) 不定期測試網站性能並進行優化與除錯。
(4.) 其他主管交辦事項。

● 研發工程師
● 韌體工程師

NEW

職務說明

(1.) 有ST MCU 開發經驗。 (2.) 有RTOS開發經驗。
(3.) 具備良好的編碼習慣，包括代碼可讀性、模組化和註釋。
(4.) 能夠獨立進行測試，具備故障排除和問題解決的能力，以確保系統的穩定性和可靠性。 (5.) 其他主管交辦事項。

需具備以下能力

(1.) STM32CubeIDE (2.) STM32CubeMX
(3.) STM32CubeProgrammer (4.) TouchGFX

Evolution
商業應用

語音功能
自然語音辨識與語音回應

導航功能
環境地圖建立與目標地點
自主規劃與前往

跟隨功能
辨識使用者的身體並跟隨

避障功能
多感測器偵測環境障礙物
自主避障與重新規畫路徑

邁克兄弟
科技股份有限公司
Smart Robot Technology Ltd.

temi 機器人

軍用公司的商用市場布局，打造AI機器人平台

temi機器人具備一套成熟先進的深度學習演算法用來偵測物體、區域、障礙物，和判斷當下環境狀況，進而讓機器人理解自己所「看」到的東西

temi機器人眞正的創新在於機器人底盤中的大腦，其透過先進的演算法，讓機器人能夠將光學雷達回傳的數據建立成精準的周遭環境2D地圖

透過光學雷達、深度鏡頭和感測器所收集到的數據，temi機可以實現流暢的自主導航與避障後的重新的路徑規劃

功能特色

1. ROBOX 自主導航系統
2. 基於 ARM 架的構導航系統，低功耗、高效率
3. 35 公尺的光學雷達掃描半徑
4. 低電量自動回充，續航力最高可達 8 小時
5. 具備 13.3“ 1080p 的 Android 平板
6. 具備 USB 孔，可外接 USB 裝置
7. 具備智慧語音互動
8. 提供 Android SDK

歡迎官網洽詢

ULTEST

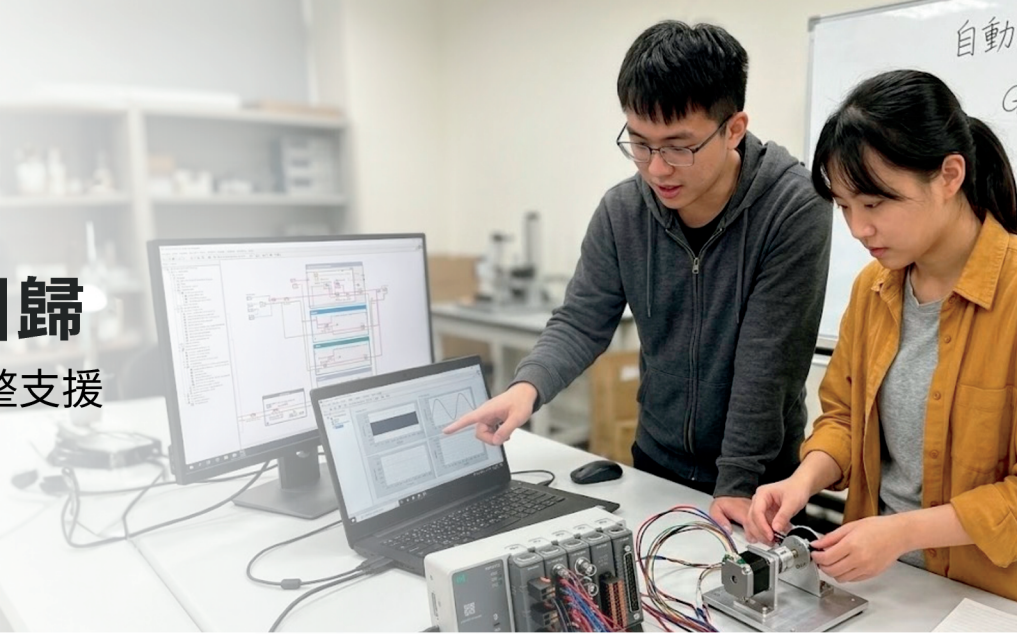
LabVIEW 學術軟體強勢回歸

一套授權，70+ NI 軟體完整支援

| 優立測科技 — NI 官方教育訓練合作夥伴

Training Partner

Authorized Distributor



研究與教學不再受限，Academic Volume License 一套授權全面支援

進階授權不足
想做 Real-Time、FPGA、HIL，卻發現單機授權工具難配置。

一套授權，70+ 完整軟體工具
包含 LabVIEW Professional、Real-Time、FPGA、Sound & Vibration、VeriStand、TestStand、DIAdem、FlexLogger、GWeb、VBAI 等軟體。

研究流程斷層
從模擬到實機，控制、量測、驗證需要不同軟體與平台？

學習不受場域限制，使用更彈性
每個 License 可安裝於 3 台個人電腦，不受特定電腦或實驗室限制，學習與研究更具彈性。

學習場域受限
研究與專題不只發生在實驗室，需要更彈性的使用方式。

一年免費線上教育訓練資源
師生可依學習進度與研究需求，一年內不限場域、無限次回看線上課程，降低學習門檻，加速技術落地。

LabVIEW AVL X Multisim AVL 軟硬體雙研發平台

一套授權，支援完整控制與硬體開發流程

LabVIEW AVL
(控制 · 量測 · AI · 自動化)

演算法 → 模擬 → 量測 → Real-Time → FPGA → HIL驗證 → AI/控制

Multisim AVL
(電路設計 · 模擬 · PCB · 原型製作)

電路設計 → 模擬 → PCB → Prototype/ Test

2026 CACS 攤位最新亮點：Nigel AI x 實機量測 DEMO

大講堂門廳 No.11

AI 不只幫你寫 Code，更能連結真實量測！

想看看 Nigel AI 如何協助新手快速上手？
透過自然語言與 Nigel AI 互動，即時獲得操作引導與排查建議，還能進行程式碼架構解析與 VI 自動生成。現場更有 Nigel AI Code Gen x CompactDAQ，快來體驗。

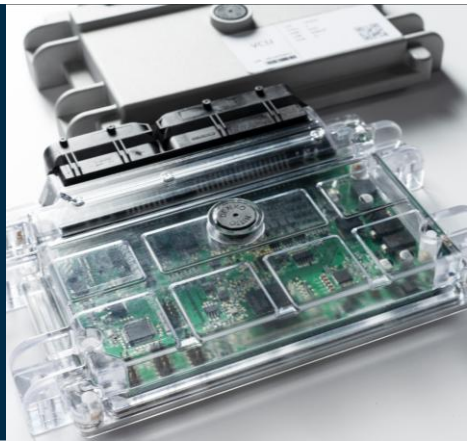
優立測科技 | NI 國家儀器台灣代理商，具備豐富學術與量測產業經驗，同時代理與服務 Iwatsu、Pickering、 DwyerOmega 、Teradyne 等國際知名品牌，滿足多元測試與控制需求。

Contact Us

VCU · AUTOSAR · AUTOMOTIVE DOMAIN AI AGENT

驅動世界的商用電動車，從軟體開始。

KopherBit 打造整車控制器（VCU）平台、AUTOSAR 工程服務，以及車用領域 AI Agent，協助電動巴士、卡車、農機與特殊車輛從概念走向量產。



01 為什麼選擇 KopherBit

軟體定義車輛（SDV）的時代已經來臨，但商用電動車製造商仍面臨開發週期冗長、工具鏈零散，以及嚴苛的功能安全與資訊安全標準。KopherBit 弭平這道落差，縮短產品上市時程，同時讓每一項工作產物皆可稽核、符合規範。

2022
工研院 Spin-off 新創團隊

15+ 年
累積車用軟體開發經驗

合規優先
ISO 26262 · ISO/SAE 21434 · ASPICE CL2

02 我們如何做到

深厚工程實力

源自工研院的團隊，橫跨動力系統、嵌入式系統與車輛 E/E 架構。

以合規為核心

工作產物遵循 ISO 26262 功能安全與 ISO/SAE 21434 資訊安全標準，可稽核、可追溯。

整合式工具鏈

從訊號設計到驗證的端到端系統，由 KDP AI 平台統一整合，建立單一可信來源。

03 我們提供什麼

VCU 控制器與硬體

KCU GEN1 / GEN2 整車控制器，以及 EVCC、T-BOX 與 BCM 模組。

AUTOSAR 與嵌入式軟體

KopherSAR CP、KopherBoot、KopherUDS、KopherV2G，搭配網路與 SWC 設計工具。

車用領域 AI 輔助設計 — KDP

橫跨整個工具鏈的 AI 輔助開發、部署、監控與訊號切換。

工具 `KITE UDS Tester` `KITE Reflasher` `KITE Calibrator` `KITE Pulse` Wind River 合作夥伴

台灣商用車領導品牌的信賴選擇
歡迎有志一同的夥伴加入！

Key Results: *As of Oct. 2025*

- 6 EV Bus Makers
- 1000+ EV on road
- >100M Mileage
- 3 Important Certification of Automotive
- 70%+ of TW E-buses

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讓我們攜手打造您的下一代電動載具控制器！

掃描 QR Code 探索完整的 VCU 與 AUTOSAR 平台，或直接與我們聯繫。

CONTACT

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yc.wang@kopherbit.com
+886-972-399-057

Pey Lu · 商務發展經理
pc.lu@kopherbit.com
+886-911-326-905

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SYNOPSYS®

TPT

從模型到實機，從需求到產品發布

一站式涵蓋完整驗證流程的平台

驗證生命週期	驗證方法	驗證流程	執行環境
MiL → SiL → PiL → HiL	黑箱測試 → 測試覆蓋率 → 故障注入	測試生成 → 測試執行 → 結果分析	本地端 → CI/CD → 雲端

涵蓋全領域應用

- 汽車
- 醫療器材
- 航太
- 國防
- 軌道運輸
- 工業
- 嵌入式軟體與系統

減少重工
降低開發後期變更與問題搶修

快速回饋
及早發現問題持續改善

降低成本
降低測試成本，提升驗證資產重用率

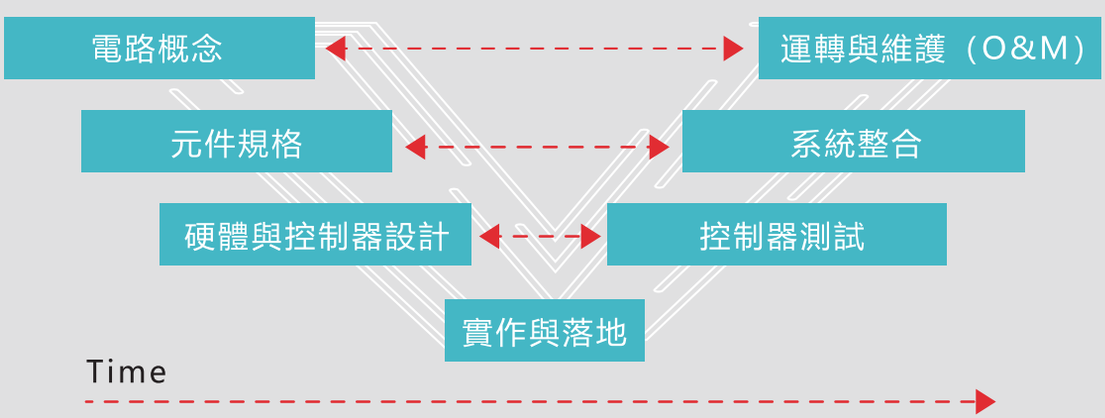
提升品質
打造可靠的軟體定義產品

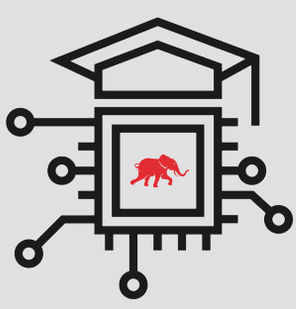
驗證邏輯貫穿全生命週期，提升重用效率

所有開發流程，盡在 Typhoon HIL Control Center

01 在 TyphoonSim 中運行模型（專案定義與前期設計階段）

02 在 HIL 模擬器上運行模型（專案測試與系統整合階段）





Typhoon HIL

為什麼選擇 Typhoon HIL?

一套模型，貫穿完整開發流程
提供學術版優惠價格
專為電力電子應用打造
無縫整合既有工具鏈

CADMEN
Taiwan Auto-Design Co. (TADC)
虎門科技股份有限公司

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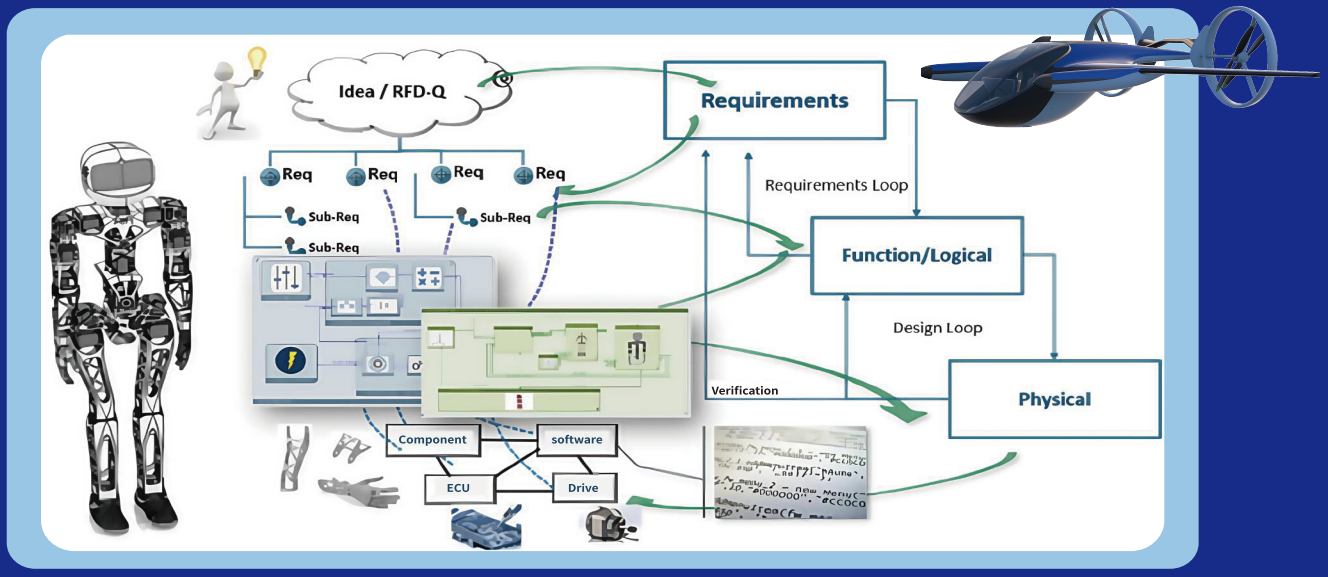
MBSE 系統工程

Model-Based Systems Engineering



在日益嚴格的監管要求與激烈市場競爭的雙重壓力下，各行各業正積極導入系統工程以應對產品日趨複雜的設計與開發挑戰。無人載具、自控機械從結構設計到實時模擬分析，產品日漸朝向高度機電整合與軟體驅動的方向發展。在此背景下，**基於模型的系統工程(MBSE)**已日益被視為設計、開發與管理此類複雜系統的關鍵任務與最佳實踐方法。

MBSE 能夠系統化地分析使用者與相關利害關係人的需求並支援設計、模擬與最佳化解決方案，涵蓋產品生命週期的各個階段 -- 從功能規劃、設計開發到驗證與最終退役。

在達梭系統的解決方案中，需求管理、系統架構設計、跨學科建模與模擬 (包含嵌入式軟體)、驗證與確認，皆整合於一個統一、協作與開放的環境。藉由端到端的可追溯性與完善的變更管理，企業能夠做出更敏捷且具洞察力的決策，同時確保符合品質與驗證標準。

CATIA系統工程

(一) 提供完整的系統建模與驗證能力

- 基於虛擬場景的測試與硬體在環(HIL)模擬
- 可重現真實環境的動態系統模型
- 以數學方式表示動態系統，並透過輸入/輸出與外部硬體或控制器交互

(二) 採用物件導向、聲明式的多領域建模語言

支援基於組件的系統建模，涵蓋機械、電氣、電子、液壓、熱力、控制、電力及流程控制等子系統間的耦合。

(三) 運用RFLP方法

運用 Requirement、Functional、Logical、Physical 進行模型分層描述與可追溯分析。這種自上而下的設計方法，能將複雜系統拆解。透過此方法，企業得以整個系統生命週期中保持結構化思維，實現跨領域整合、設計最佳化與高可靠性。

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